GRACE / GRACE-FO SCIENCE TEAM MEETING 2026

GSTM 2026

Abstract Book · 91 contributions

POTSDAM, GERMANY

GFZ Helmholtz Centre for Geosciences

13–15 October 2026 · Hybrid

COMPILED

2026-10-28 · 95 submitted abstracts, 4 withdrawn

Source: meetingorganizer.copernicus.org/GSTM2026

DOIs: 10.5194/gstm2026-*

Licence: abstracts © authors · CC BY 4.0

Contents

1. GSTM2026-1 · FEATS-G: Orbit and quantum sensor performance simulation for future satellite gravity missions
2. GSTM2026-2 · Learning from ocean bottom pressure differences between GRACE and a state estimate constrained by GRACE
3. GSTM2026-4 · Towards hydrological extreme event monitoring using deep learning-downscaled simulated NGGM and MAGIC data and GRACE(-FO) observations
4. GSTM2026-6 · Testing mascon regularization strategies for simulated future gravity-field recovery
5. GSTM2026-7 · Multiyear tropical warm pool warming drives slowdown in Antarctic mass loss
6. GSTM2026-8 · Antarctic and Greenland Ice Sheet Gravimetric Mass Balance from GRACE(-FO): TU Dresden's ESA CCI+ Products and Component Partitioning for Antarctica
7. GSTM2026-9 · GRACE and satellite altimetry to measure ocean mass and density changes between 2002 and 2026
8. GSTM2026-10 · Uncovering the Onset of Water Storage Droughts in the Danube River Basin
9. GSTM2026-12 · Machine Learning-Based Accelerometer Transplantation for GRACE and Future Quantum-Enabled Satellite Gravimetry
10. GSTM2026-13 · When and where does groundwater recharge occur in arid lands?
11. GSTM2026-14 · Impact of Heterogeneous Mantle Viscosity on the Glacial Isostatic Adjustment Correction for GRACE(-FO): Implications for Surface Mass Trends
12. GSTM2026-15 · ESA-SING — Simulated products for future gravity missions
13. GSTM2026-16 · Ocean mass changes from the lower degree harmonics of the GRACE/GFO monthly gravity fields.
14. GSTM2026-17 · From tight to wide deadband: where current model-based GRACE-FO transplants fall behind
15. GSTM2026-18 · Advancing glacier and ice-sheet mass-change observations with GRACE-C, NGGM and MAGIC
16. GSTM2026-19 · Arctic glacier mass change from GRACE/GRACE-FO: Regional variability, atmospheric drivers and rapid GIA
17. GSTM2026-20 · Using GRACE-/FO 10-day CNES solutions for hydrological applications
18. GSTM2026-21 · Developing the SAGSA L3 gravimetry ensemble and its applications
19. GSTM2026-22 · The satellite-only gravity field model GOCO2025s
20. GSTM2026-23 · Impact of the new GFZ RL07 GRACE/GRACE-FO processing standards on the Level-2 gravity field solutions
21. GSTM2026-24 · Future Satellite Gravimetry for Global Total Drainable Water Storage Estimation: Mission Performance and Basin Controls
22. GSTM2026-25 · COST-G: new options and motivation for gravity field combinations on normal equation level
23. GSTM2026-26 · Unified Recovery of Low-Degree Time-Variable Gravity and Terrestrial Reference Frame Parameters Through Multi-Technique Space Geodesy
24. GSTM2026-27 · Insights from the DANSk-LSM Project to Advance GRACE(-FO) and Multi-Sensor Data Assimilation for Hydrological Monitoring and Early Warning Systems
25. GSTM2026-28 · JPL GRACE and GRACE-FO Level-2 Overview
26. GSTM2026-29 · GRACE Data as Non-Tidal Deformation in Reference Frame Determination: From Position Timeseries Forecast to Frame Realization
27. GSTM2026-30 · Leveraging Satellite Altimetry and Gravimetry for High-Resolution Antarctic Mass Change
28. GSTM2026-31 · Interaction of Deterministic Errors in GRACE-Continuity
29. GSTM2026-32 · Ellipsoidal Equal-Area Panel Mascons from GRACE Intersatellite Geopotential Differences: A Preliminary Assessment over China
30. GSTM2026-33 · Data-driven Uncertainty-aware 5-day Surface Mass Variations from GRACE Ranging Residuals
31. GSTM2026-34 · Downscaling and Gap-Filling GRACE-Based Terrestrial Water Storage Anomalies Using Bias-Aware Machine Learning
32. GSTM2026-35 · MicroSTAR: NGGM Space Accelerometer Last Developments
33. GSTM2026-36 · Mass column height determined from a TVG field by inversion in the spatial domain
34. GSTM2026-37 · Assessing the potential of future gravity missions to monitoring the Atlantic Meridional Overturning Circulation
35. GSTM2026-38 · Directional gradient regularization of gravity field solutions
36. GSTM2026-39 · Optimized SLR mass change and degree 1 estimation with a genetic algorithm
37. GSTM2026-40 · ODIN and THOR: optomechanical inertial sensors for space geodesy
38. GSTM2026-41 · Deployment of a Long-Term Self-Calibrating Ocean Bottom Pressure Recorder in the Nordic Seas for GRACE-FO, GRACE-C, and NGGM Validation
39. GSTM2026-42 · Enhanced Retrieval of the Seasonal Cycle of Terrestrial Water Storage
40. GSTM2026-44 · Stochastic modeling of non-tidal atmospheric and oceanic dealiasing models for GFZ GRACE/GRACE-FO Level-2 processing
41. GSTM2026-45 · Ground experiment to study the interaction of a bounded plasma with a metal shield
42. GSTM2026-46 · The Response of GRACE/GRACE-FO to 14,000 Global Lakes from 1992 to 2026
43. GSTM2026-47 · Quantifying total drainable water storage and its distribution using GRACE/GRACE-FO
44. GSTM2026-48 · GRACE/GRACE-FO radio occultations for weather forecast and climate research
45. GSTM2026-49 · Time-Variable Gravity Recovery from Clock Frequency Comparisons in Formation-Flying Missions
46. GSTM2026-50 · Enhancing the Spatial Resolution of Monthly Gravity Solutions Derived from SLR and DORIS
47. GSTM2026-52 · Localized Estimation of Tide Model Error and Mass Anomalies from GRACE/GRACE-FO Inter-Satellite Ranging
48. GSTM2026-53 · Spatiotemporal Validation of Mass-Change Products over Greenland and Antarctica Using GRACE-FO Laser Ranging Instrument
49. GSTM2026-54 · Assessing Machine Learning for Automated Screening of GRACE-C Quicklook Gravity Solutions
50. GSTM2026-55 · Seismic Effects of Large Earthquakes on Sea-Level Budget Closures
51. GSTM2026-56 · Sub-monthly Terrestrial Water Storage Change from GRACE-C and NGGM: Assessing MAGIC's Capability for Rapid Mass Change Monitoring
52. GSTM2026-57 · Is the contribution of glaciers and ice sheets to sea level rise slowing down?

53. GSTM2026-58 · GRACE Reveals Contrasting Groundwater Responses to Surface-Water Shortages Across the Colorado River Basin
54. GSTM2026-59 · Observing Long-Term Mass Changes in the Global Ocean and Ice Sheets from Satellite Gravimetry: 1993–2023
55. GSTM2026-60 · A Framework for the Assimilation of GRACE/GRACE-FO Observations into JPL Terrestrial Reference Frame Products
56. GSTM2026-61 · Hydrological de-aliasing model data for GRACE/-FO gravity field processing
57. GSTM2026-62 · The updated ESA Earth System Model for Future Gravity Mission Simulation Studies: ESA ESM 3.0
58. GSTM2026-63 · Beyond GRACE: Benchmarking Global Hydrological Models Back to the 1990s with Satellite Laser Ranging
59. GSTM2026-64 · Assimilation of GRACE/-FO data into global hydrological models for water use scenarios
60. GSTM2026-65 · Using GRACE as an Observational Constraint for Exponential Depth Extrapolation of Satellite Soil Moisture
61. GSTM2026-66 · Signal separation in GRACE and GRACE-FO data using a deep learning algorithm
62. GSTM2026-67 · Stochastic modelling in attitude determination of GRACE-FO satellites for ITSG gravity field solutions
63. GSTM2026-68 · A traceable GRACE/GRACE-FO Level-1C scientific dataset (2002-2026) for reproducible mass-change analysis
64. GSTM2026-69 · Benefits of future satellite gravimetry missions for characterizing extreme wet events in terrestrial water storage
65. GSTM2026-70 · From Development to Systematic Production: SAGSA Generates Space-Based Gravimetry Products (L2–L4) from GRACE and GRACE-FO Data for the Scientific Community
66. GSTM2026-71 · GRACE/-FO Data Assimilation for Hydrological Applications: From Model-Specific Solutions Towards Broader Use
67. GSTM2026-72 · Evaluation of the OMP 2025 AOD1B dealiasing product based on the TUGO ocean dynamics numerical model
68. GSTM2026-73 · A detailed description of CNES monthly and 10-day time-variable gravity solutions and a summary of the improvements to be applied to the next reprocessing
69. GSTM2026-74 · COM-TG: A combined mascon product from multiple analyses of GRACE and GRACE Follow-On space gravity data
70. GSTM2026-75 · Inter-comparison of Greenland-wide annual mass trends estimated with the input-output method and extracted from GRACE level-2 data
71. GSTM2026-77 · Preliminary results of the GFZ daily GRACE/GRACE-FO Kalman filter solution
72. GSTM2026-78 · A Physics-Informed Dual-Stream Architecture for Spatiotemporal Downscaling of GRACE/GRACE-FO to Daily Resolution
73. GSTM2026-79 · A GRACE-based framework for estimating specific yield at regional scale across India
74. GSTM2026-80 · Assessment of continental water mass balance components 2003--2022 over EURO-CORDEX
75. GSTM2026-81 · Towards a SmallSat-compatible Laser Ranging Instrument for Future Multi-Pair Constellations
76. GSTM2026-82 · Precision of Antarctic mass balance estimates from time-variable gravity limited by atmospheric pressure uncertainties
77. GSTM2026-83 · Status of the GRATTIS Mission
78. GSTM2026-84 · GFZ SDS Level-2/-3 Overview
79. GSTM2026-85 · Mantle Rheology Across Tidal to Decadal Timescales
80. GSTM2026-86 · GRACE Follow-On LRI measurements of linear accelerations produced by attitude control thrust firings
81. GSTM2026-87 · Attribution of spatially refined terrestrial water storage trends to human activities
82. GSTM2026-88 · GRACE-C Implementation Status - Update
83. GSTM2026-90 · GRACE and GRACE-FO Level-1 Data Processing Status
84. GSTM2026-91 · GRACE-FO Mission Operations Report
85. GSTM2026-92 · GRACE/GRACE-FO Observations Reveal Long-Term Groundwater Depletion and Freshwater Storage Losses in the Colorado River and Rio Grande-Bravo Basins
86. GSTM2026-95 · TN-14: Validation efforts and impacts of AOD1B RL07
87. GSTM2026-96 · Status of AOD1B
88. GSTM2026-97 · From GRACE Science to Climate Service: Operational TWSA and GWSC Products in the Copernicus Climate Change Service (C3S)
89. GSTM2026-99 · CSR GRACE and GRACE-FO Level-2/-3 Overview
90. GSTM2026-101 · GRACE-FO: Science Management Report
91. GSTM2026-103 · GRACE-FO: mission status, science performance and outlook

GSTM2026-1 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · POSTER

FEATS-G: Orbit and quantum sensor performance simulation for future satellite gravity missions

Annike Knabe1, Henrik Wessel2, Stefanie Bremer2, and Matthias Weigelt1

German Aerospace Center (DLR), Institute for Satellite Geodesy and Inertial Sensing, Hannover, Germany · German Aerospace Center (DLR), Institute for Satellite Geodesy and Inertial Sensing, Bremen, Germany

Satellite gravimetry missions GRACE and GRACE-FO have significantly advanced Earth observation by enabling monthly, high-precision mapping of global mass redistributions. These missions have provided unprecedented insights into the system Earth, such as ice sheet and glacier mass loss, sea level rise, terrestrial water storage changes. However, the classical electrostatic accelerometers suffer from their low-frequency drift and are one of the limiting factors. This drawback can be counterbalanced by an accelerometer based on Cold Atom Interferometry (CAI) because of its high long-term stability.The FEATS-G project advances this frontier through end-to-end simulations to evaluate quantum-based satellite mission concepts. The simulations allow for a comprehensive evaluation of future mission concepts and the exploration of novel scientific applications. They also enable the assessment of quantum-based sensors' added value in distinguishing between different geophysical signals, while preparing the scientific community for the arrival of this new data type.This contribution focuses on the simulation procedure of observations based on quantum technology. The selected mission scenario will be simulated by the orbit propagation software VENQS. The satellites' surface is modelled by means of finite elements in order to calculate precise non-gravitational forces induced by solar and Earth radiation and atmospheric drag. The non-gravitational accelerations are considered as disturbance and are measured by the accelerometer. The CAI acceleration measurements are simulated considering various error sources, such as detection noise, contrast loss, wavefront aberration and laser frequency noise. Furthermore, rotational effects and rotation compensation methods using a counter-rotating mirror are included. The simulation procedure and methodology and first results are shown. The artificial orbit data and sensor output will later be used in the gravity field recovery process.We acknowledge the support by the European Union for the project FEATS-G (Call: HORIZON-CL4-2025-02-SPACE-81).

DOI: 10.5194/gstm2026-1 · https://doi.org/10.5194/gstm2026-1

GSTM2026-2 B.3 · OCEANOGRAPHY · ORAL

Learning from ocean bottom pressure differences between GRACE and a state estimate constrained by GRACE

Rui Ponte1, Nishchitha Silva1, Ou Wang2, Ichiro Fukumori2, and Mengnan Zhao1

Atmospheric and Environmental Research, Lexington, MA, United States of America · Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States of America

Observations from GRACE and its follow-on mission have been used to constrain the ocean bottom pressure (OBP) fields in the state estimates produced by the project for Estimating the Circulation and Climate of the Ocean (ECCO). Here we examine the OBP residuals, i.e., the remaining differences between a recent optimized ECCO solution and the GRACE data, to identify potential issues with both the observations and the ocean model used in ECCO. The GRACE-ECCO residuals are dominated by long-term trends and seasonal variability and exhibit mostly large-scale structures. The trends can be partly related to unmodeled earthquake signals and gravitational, rotational and deformational (GRD) effects on the GRACE-derived OBP fields. Land hydrology processes and associated GRD effects can also contribute to seasonal residuals. In regions of high eddy energy (e.g., Agulhas Retroflection, Gulf Stream), residuals are partly caused by the impact of mesoscale signals on GRACE-derived OBP fields, which is not represented in the coarse-resolution ECCO model. Other possible contributions to the residuals include uncertainties in glacial isostatic adjustment models and poor horizontal resolution in both data (leakage of land signals in GRACE retrievals) and model (inadequate representation of coastal domains).

DOI: 10.5194/gstm2026-2 · https://doi.org/10.5194/gstm2026-2

GSTM2026-4 B.4 · HYDROLOGY · ORAL

Towards hydrological extreme event monitoring using deep learning-downscaled simulated NGGM and MAGIC data and GRACE(-FO) observations

Gilberto Goracci1, Celina Anael Farias2,3, and Ilias Daras4

ESA Φ-lab Explore Office (EOP-S8E) - Climate Action, Sustainability and Science Dept. – Earth Observation Programmes Directorate · Sapienza University of Rome · TITAN4 · ESA Earth and Mission Sciences Division - Climate Action, Sustainability and Science Dept. – Earth Observation Programmes Directorate

Terrestrial Water Storage Anomalies (TWSA) derived from satellite gravimetry provide a unique, spatially integrated measure of large-scale hydrological conditions and are widely used to characterize basin-scale wet and dry states. However, their applicability to regional hydrological monitoring remains limited by their coarse spatial resolution, which hampers the detection of localized storage anomalies relevant to flood onset, evolution, and groundwater variability.

Here, it is investigated an unsupervised deep-learning framework based on convolutional neural networks to enhance the spatial resolution of TWSA products across simulated and real mission scenarios. The method relies on a U-Net architecture driven by ERA5-Land hydro-climatic variables, and is first applied to end-to-end closed-loop simulations of the NGGM and MAGIC next-generation gravity missions. Particular attention is given to 5-daily solutions, which represent a key expected advancement of future gravimetric constellations compared with the monthly solutions currently provided by GRACE(-FO). In this setting, simulated TWSA fields are downscaled from 3° to 1°, while monthly 2° simulations are also considered to assess the framework under GRACE-like temporal conditions. For the syntetic products, the downscaled fields are validated using the ESA Earth System Model 3.0 as the reference signal.

Results show that the downscaled products preserve consistency with native gravimetric observations while recovering finer-scale spatial structures. To assess transferability, the same pipeline is applied to real GRACE(-FO) observations, downscaling monthly 3° TWSA products into 1° maps.The framework is also extended to GRACE-derived groundwater storage anomalies (GWSA) over France, downscaled from 0.5° to 0.1° and evaluated against independent in situ groundwater observations.Finally, downscaled synthetic TWSA fields are exploited for extreme-event monitoring using threshold-based indicators derived from climatological anomalies. The analysis assesses whether the increased spatial resolution retains localized anomaly patterns otherwise smoothed at coarse scales, supporting the monitoring of flood-prone conditions.

DOI: 10.5194/gstm2026-4 · https://doi.org/10.5194/gstm2026-4

GSTM2026-6 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · POSTER

Testing mascon regularization strategies for simulated future gravity-field recovery

Josefine Wilms1, Christoph Dahle1, Markus Hauk1, Natalia Panafidina1, Michael Murboeck1,2, and Annette Eicker1

Deutsches GeoForschungsZentrum GFZ, Section 1.2: Global Geomonitoring and Gravity Field, Weßling, Germany · Technische Universität Berlin, Fakultät VI Planen Bauen Umwelt, Institut für Geodäsie und Geoinformationstechnik, Berlin

Mascon solutions estimate mass changes directly on predefined surface elements, rather than only representing them by spherical harmonic coefficients. They are useful for regional water storage, ice-mass, and ocean applications, and are already provided by several groups such as GSFC, JPL and CSR. In the framework of GRACE-C Science Data System preparations at GFZ, a project funded by DLR, we are developing and testing mascon processing strategies using realistic full-noise simulations of monthly gravity-field solutions.

We first consider a GRACE-C-like single polar-pair scenario, representing the continuation of the US-German GRACE mission series after GRACE and GRACE-FO. We then compare this with the proposed MAGIC double-pair constellation, an ESA-NASA concept in which the NASA/DLR GRACE-C polar pair is complemented by ESA's

inclined NGGM pair. By comparing the two scenarios, we investigate how the observation geometry affects the recovered mascon fields and the required regularization.

The processing starts from monthly spherical harmonic solutions. These are regularized in coefficient space before being transformed to the mascon grid. We test different spherical-harmonic regularization settings and inspect their effect on striping, ocean RMS, and land signal preservation.

In the mascon step, we compare two ways of building the regularization matrix. The first uses an intermediate mascon solution to construct a time-variable regularization matrix. The second derives the time-variable regularization matrix directly from synthesized spherical harmonic fields, avoiding the intermediate mascon step.

The recovered fields are compared with the known input signal to evaluate noise reduction and signal preservation. Preliminary results show that time-variable mascon regularization can reduce errors compared with the first mascon solution, but the best balance between noise reduction and signal preservation is still being evaluated. This work presents the current workflow, first results, and open questions for discussion.

DOI: 10.5194/gstm2026-6 · https://doi.org/10.5194/gstm2026-6

GSTM2026-7 B.2 · CRYOSPHERE · ORAL

Multiyear tropical warm pool warming drives slowdown in Antarctic mass loss

Qinghua Ding, Yunhe Wang, Xiaofeng li, Thomas Ballinger, Yoshihiro Nakayama, Danial Topal, and Eric Steig

University of California Santa Barbara, Santa Barbara, United States of America University of California Santa Barbara, Santa Barbara, United States of America

Antarctic mass loss has been a major contributor to global sea-level rise for most of the last few decades, mainly driven by West Antarctica1. During 2021–2023, however, a sharp increase in surface mass balance over Queen Mary Land and Wilkes Land in East Antarctica offset West Antarctic loss and slowed the rate of total ice mass loss2,3. Although this slowdown is consistent with the expected long-term precipitation response to global warming through poleward-shifted storm tracks and Antarctic moistening4, our results point to a different mechanism. Here we show that the recent ice mass gain was linked to a recurrent atmospheric teleconnection driven by sea surface temperature anomalies in the tropical warm pool, which experienced unusually persistent warming from 2021 to 2023 relative to the previous two decades. On the basis of observations and model experiments, we find that tropical warm pool warming excites a poleward-propagating Rossby-wave train that induces a high-pressure anomaly over East Antarctica, enhancing Queen Mary Land and Wilkes Land precipitation and driving the observed mass gain, with moisture primarily sourced from the mid-latitude Indian Ocean. Similar multiyear warming in the tropical warm pool recurs about once per decade in observations and historical simulations, and its influence on precipitation is distinct from the effects of global warming. Therefore, the recent Antarctic Ice Sheet mass gain is probably temporary and does not yet reflect a sustained, global-warming-driven moistening of Antarctica.

DOI: 10.5194/gstm2026-7 · https://doi.org/10.5194/gstm2026-7

GSTM2026-8 B.2 · CRYOSPHERE · POSTER

Antarctic and Greenland Ice Sheet Gravimetric Mass Balance from GRACE(-FO): TU Dresden’s ESA CCI+ Products and Component Partitioning for Antarctica

Alexey Kupriyanov, Martin Horwath, Matthias O. Willen, and Thorben Döhne

TUD Dresden University of Technology, Institute of Planetary Geodesy, Germany TUD Dresden University of Technology, Institute of Planetary Geodesy, Germany

For more than two decades, the Gravity Recovery and Climate Experiment (GRACE) and its follow-on mission GRACE-FO have been providing unique information on mass changes occurring globally and in the Antarctic (AIS) and Greenland (GIS) ice sheets particularly. Ice sheets were assigned by the Global Climate Observing System (GCOS) as one of the essential climate variables. GRACE(-FO) observations showed that both AIS and GIS have an

accelerating ice mass loss. Moreover, ice sheets are considered as one of the largest sources of uncertainty in future global sea-level rise projections. Therefore, consistent analysis of the ice sheet mass changes is needed for improving the understanding of their contribution to the global sea-level rise.We present the recently published extension (through December 2025) of the gravimetric mass balance (GMB) gridded and drainage basins products by TU Dresden for the AIS and GIS that was done in the framework of the European Space Agency's (ESA) Climate Change Initiative (CCI) projects. The products are based on the monthly GRACE(-FO) solutions CSR RL06.3 by the Center for Space Research at the University of Texas at Austin with applied tailored sensitivity kernel approach, developed at TU Dresden (Groh and Horwath 2021). For Antarctica a novelty of this release is a gap-filling interpolation for the 11-months GRACE/GRACE-FO transition period using the Regional Atmospheric Climate Model RACMO2.4p1. For the investigated period of more than 23 years, the GMB of the AIS was quantified to be −83±44 Gt a-1 and for GIS −246±14 Gt a-1, that corresponds to a mean sea-level rise of 0.23±0.12 mm a-1 and −0.68±0.04 mm a-1 respectively.The total mass change observed by GRACE(-FO) is the combined effect of variations in surface mass balance (SMB) changing ice-flow dynamics. The Partitioned GMB (GMBP) product is being developed to reflect these two physically distinct components. SMB variation effects are predominantly short-term or interannual and are controlled by ice-atmosphere interaction. Ice-flow dynamics effects, in contrast, are mostly long-term (decadal or longer), controlled by ice-ocean interaction and most crucial for understanding long-term evolution of the ice sheet. We present preliminary methods and results of the Partitioned GMB.

DOI: 10.5194/gstm2026-8 · https://doi.org/10.5194/gstm2026-8

GSTM2026-9 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

GRACE and satellite altimetry to measure ocean mass and density changes between 2002 and 2026

Ernst Schrama and Wouter van der Wal

TU Delft, Faculty of Aerospace Engineering, Space Department, Delft, Netherlands TU Delft, Faculty of Aerospace Engineering, Space Department, Delft, Netherlands

From GRACE and GRACE-FO we get time variable information from the Earth’s gravity field since 2002. The monthly time series obtained from science centers can be represented as a surface water layer. By sampling the oceans at mascons we obtain a mass against time signal which represents the barystatic component of the sea level rise signal which is confronted with satellite altimetry since 1992, the combination of both geodetic results allows to estimate the density change in the ocean domain for which profiling float measurements can be used as an independent source. Of particular interest is the effect of glacial isostatic adjustment which shows up in both gravimetry and altimetry. The GIA correction determines for a part the implementation of the geocenter correction that is used during the GRACE data processing. Depending on the processing choices to represent ocean mass and the GIA correction that is required we find barystatic sea level rates varying between 1.9 and 2.4 mm/yr whereby we remark that the uncertainty is larger than previously expected. The volume change in the oceans is remarkably stable, it remains at 3.7 mm/yr with a noticeable acceleration of 0.2 mm/yr2. This means that the density change due to temperature and salinity changes in the ocean domain varies between 1.3 and 1.8 mm/yr.

DOI: 10.5194/gstm2026-9 · https://doi.org/10.5194/gstm2026-9

GSTM2026-10 B.4 · HYDROLOGY · ORAL

Uncovering the Onset of Water Storage Droughts in the Danube River Basin

Eva Boergens1, Sangita Chouhan1, and Julian Haas2

GFZ Helmholtz Centre for Geosciences, Sec. 1.3 Earth System Modelling, Potsdam, Germany · GFZ Helmholtz Centre for Geosciences, Sec. 4.4 Hydrology, Potsdam, Germany

Over the past decade, Europe has experienced several interconnected droughts. Since 2018, TWS has shown no storage surplus across the majority of Central Europe. However, does this truly mark the onset of these drought

conditions, or did they start earlier? To answer this question, we apply the BFAST (Breaks For Additive Season and Trend) method to detect breakpoints in the TWS time series, aiming to pinpoint the onset of the ongoing drought. Here, we focus on the Danube River Basin. Since TWS represents the sum of all water storage compartments (WSCs) — groundwater storage, surface water storage, root-zone soil moisture, snow water equivalent, and glacier storage — we further investigate datasets for each individual WSC. Since hydro-meteorological fluxes, namely precipitation and evapotranspiration, are the significant drivers of TWS changes, we also include relevant hydro-meteorological indices (SPI, SPEI, EDDI) in the study.BFAST identifies a significant breakpoint in TWS as early as 2015, and an even earlier one in groundwater storage in 2013, while soil moisture remains stable throughout. Investigation of the hydro-meteorological drivers of storage change reveals that precipitation remained stable over the full study period (2002-2025), and that the observed decline in water storage is instead attributable to increased evapotranspiration driven by rising temperatures.Further analysis confirms the significance of the 2015 breakpoint in TWS: storage was stable beforehand but has declined significantly since. The temporal behaviour of storage is examined further using a transition matrix and a duration frequency analysis. Prior to 2015, TWS remained predominantly in a normal state, whereas after 2015 it exhibited more frequent and prolonged dry periods. These results confirm the significance of the ongoing shift in the hydrological system of the Danube River Basin in recent years.

DOI: 10.5194/gstm2026-10 · https://doi.org/10.5194/gstm2026-10

GSTM2026-12 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Machine Learning-Based Accelerometer Transplantation for GRACE and Future Quantum-Enabled Satellite Gravimetry

Mohsen Romeshkani, Jürgen Müller, and Sahar Ebadi

Leibniz University Hannover, Institut für Erdmessung, Hannover, Germany Leibniz University Hannover, Institut für Erdmessung, Hannover, Germany

Accurate and continuous determination of the Earth's time-variable gravity field is essential for monitoring mass redistribution related to hydrological, climatic, and geodynamic processes. GRACE and GRACE Follow-On (GRACE-FO) have demonstrated the high capability of low-low satellite-to-satellite tracking (LL-SST) technique for global gravity field recovery; however, the resulting accuracy strongly depends on the quality of the involved sensors, e.g., the accelerometers (ACC) used to account for non-gravitational forces. While conventional electrostatic accelerometers (EA) provide high short-term sensitivity, their performance is affected by drift at low frequencies. Emerging quantum-based sensors, particularly Cold Atom Interferometry (CAI) accelerometers, offer promising characteristics to overcome the later issue for future satellite gravimetry.

This study investigates accelerometer observation transplantation to reduce the number of accelerometers required onboard future satellite gravimetry missions. Different configurations involving EA, CAI, and hybrid EA–CAI accelerometers are evaluated using closed-loop LL-SST simulations. Four scenarios are considered, ranging from the conventional EA-only configuration to an ideal configuration in which both satellites carry complete hybrid accelerometers. Particular attention is given to a reduced-sensor configuration in which EA and CAI observations are transplanted between satellites, enabling a virtual hybrid configuration with fewer physical accelerometers.

In addition to conventional analytical transplantation, machine-learning (ML) approaches, including Multilayer Perceptron (MLP), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM), are investigated to improve the transplanted accelerometer observations. The configurations are evaluated directly at the gravity-field level. The results show that the complete dual-hybrid configuration provides the highest performance, while the ML-based reduced-sensor configuration significantly improves upon analytical transplantation and approaches the performance of the ideal hybrid configuration.

The data-driven transplantation concept is further assessed using real GRACE accelerometer observations. During GRACE and GRACE-FO, accelerometer degradation or failure affected the availability and quality of non-gravitational acceleration observations, requiring their reconstruction. An LSTM-based dynamic mapping using GRACE-A accelerometer observations and auxiliary non-gravitational force models is developed to reconstruct

GRACE-B accelerations. The reconstructed observations are introduced into monthly gravity field recovery and evaluated through degree variances, equivalent-water-height representations, and RMS variability over the oceans. The results demonstrate successful reconstruction of the missing accelerometer information and improvements in the resulting monthly gravity field solutions. The same framework can also be adapted to degraded accelerometer observations in GRACE-FO.

Overall, the results demonstrate the potential of ML for recovering missing or degraded accelerometer observations and enabling reduced-sensor architectures for future satellite gravimetry. Combining quantum sensing, accelerometer transplantation, and ML provides a promising pathway toward high-performance, less complex, and potentially more cost-effective future gravimetry missions.

Keywords: Satellite gravimetry; GRACE; GRACE-FO; accelerometer transplantation; Cold Atom Interferometry; hybrid accelerometers; machine learning; time-variable gravity field.

Acknowledgement: This work was supported by Germany's Excellence Strategy – EXC-2123 "QuantumFrontiers" – 390837967.

DOI: 10.5194/gstm2026-12 · https://doi.org/10.5194/gstm2026-12

GSTM2026-13 B.4 · HYDROLOGY · ORAL

When and where does groundwater recharge occur in arid lands?

Mohamed Sultan, Hassan Saleh, and Rami W Mansouri

Western Michigan University, Department of Geological and Environmental Sciences, KALAMAZOO, United States of America Western Michigan University, Department of Geological and Environmental Sciences, KALAMAZOO, United States of America

Isotopic and geochronologic evidence indicates that many aquifers in arid lands were recharged during past wetter climatic periods, yet where and when modern recharge occurs remains poorly constrained. By integrating GRACE and GRACE-FO terrestrial water storage (TWS) observations, GPM precipitation data, groundwater-level measurements, and groundwater isotopic compositions, we documented aquifer recharge following tropical cyclones and extreme precipitation events in Oman's karstic aquifers and the fractured-basalt aquifers of western Arabia. We then extended the analysis across nine geographic domains defined by continental location and the boundaries of the major arid zones. Our results show that extreme precipitation events generated groundwater storage responses detectable by GRACE across approximately two-thirds of arid regions, with recharge efficiencies exceeding conventional estimates based on long-term averages of precipitation and recharge. These findings indicate that GRACE TWS records carry more information about episodic groundwater recharge than previously recognized. The forthcoming GRACE-C mission will help minimize gaps in the monthly TWS record, and the Next Generation Gravity Mission (NGGM) will provide enhanced spatial and temporal resolution for refining estimates of modern aquifer recharge.

DOI: 10.5194/gstm2026-13 · https://doi.org/10.5194/gstm2026-13

GSTM2026-14 B.1 · SOLID EARTH SCIENCES · ORAL

Impact of Heterogeneous Mantle Viscosity on the Glacial Isostatic Adjustment Correction for GRACE(-FO): Implications for Surface Mass Trends

Erin Hightower1, Lambert Caron1, Felix Landerer1, Michael Watkins2,3, Erik Ivins1, and Eric Larour1

Solid Earth and Ice Group, Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA USA · Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, CA, USA · Division of Engineering and Applied Science, California Institute of Technology, Pasadena, CA, USA

Accurate interpretation of time-variable gravity from GRACE(-FO) requires correction for glacial isostatic adjustment (GIA), which has a gravitational signal similar in magnitude to that of contemporary hydrological and cryospheric mass change. Traditional GIA models used to correct space gravimetry data assume a radially symmetric mantle

viscosity structure, ignoring the 3D structure evident in seismic tomography. We quantify the sensitivity of GRACE-derived surface mass trends to laterally heterogeneous mantle viscosity by computing a suite of global GIA models using the finite-element code CitcomSVE. We utilize 3D viscosity fields derived from tomography and two commonly used ice-loading histories, systematically increasing heterogeneity about their associated radial viscosity profiles. GIA solutions are converted to GRACE-compatible Stokes coefficients and equivalent water height to recompute GRACE mass trends. While global mean ocean and Greenland ice mass trends are weakly affected, regional terrestrial water storage and ocean mass patterns exhibit large and spatially coherent changes. 3D GIA corrections lead to significantly more drying conditions in Canada and slightly more drying conditions throughout most land-masses, can lead to a reduction in the long-term rate of ice mass loss in Antarctica, and can reverse the ocean mass trend in key ocean basins. Models combining elevated lower-mantle viscosity with strong lateral heterogeneity eliminate problematic GRACE ocean anomalies in the eastern North Atlantic and potentially resolve the disagreement between GIA- and geodynamics-derived estimates of mantle viscosity. Our results demonstrate that realistic 3D viscosity is essential for quantifying and reducing GIA-related uncertainty in GRACE interpretations and can ultimately lead to improved confidence in the inferred surface mass changes.

DOI: 10.5194/gstm2026-14 · https://doi.org/10.5194/gstm2026-14

GSTM2026-15 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · POSTER

ESA-SING — Simulated products for future gravity missions

Marius Schlaak1, Julia Pfeffer2, Roland Pail3, Marie Bouih2, Alejandro Blazquez1, Benoit Meyssignac1, and Ilias Daras4

LEGOS (CNES, CNES, IRD, UT), Toulouse, France · Magellium, 31520, Ramonville Saint-Agne, France · Technical University of Munich, TUM School of Engineering and Design, Chair for Astronomical and Physical Geodesy, Munich, Germany · ESA/ESTEC, Noordwijk, Netherlands

The ESA-SING dataset has been created within the ESA-SING project to investigate current and future satellite gravity mission concepts. These synthetic observations support mission-impact studies within the ESA-SING project, evaluating the added value of the NGGM and MAGIC missions for scientific applications and operational services in hydrology, ocean sciences, glaciology, climate sciences, solid earth sciences, and geodesy.The ESA-SING synthetic observations are based on input models representing global mass changes over a 12-year period available from the ESA Earth System Model, incorporating realistic instrument errors and background uncertainties for tidal and non-tidal de-aliasing products. The uncertainties in simulated single polar pair missions are comparable to those in historical GRACE records, as confirmed by the consistency in barystatic sea level errors between GRACE and GRACE-C-like data.The ESA-SING dataset contains a 12-year time series of 30-day and 5-day estimates of the time-variable gravity field, together with estimates of the long-term trend and annual cycle, representing the ESM-simulated mass transport in the hydrosphere, cryosphere and Solid Earth and incorporating realistic uncertainties for each mission configuration. Simulated satellite gravimetry products are available in spherical harmonics (L2) and equivalent water height (L3) grids for three mission concepts: a single inline-pair in polar orbit (GRACE-C-like), a single pair in an inclined orbit (NGGM), and the combination of both (MAGIC). By evaluating the recoverability of time-variable mass signals across multiple temporal scales, The ESA-SING dataset yields representative error estimates for all three satellite constellations.This work provides an overview of the SING simulations and assesses the performance of the satellite configurations considered, demonstrating that the future inclined pair (NGGM) and double-pair (MAGIC) concepts reduce noise levels in time-variable gravity field retrievals compared to single-pair observations. Moreover, this contribution will provide an introduction to the ESA-SING dataset to assist potential users.

DOI: 10.5194/gstm2026-15 · https://doi.org/10.5194/gstm2026-15

GSTM2026-16 B.3 · OCEANOGRAPHY · POSTER

Ocean mass changes from the lower degree harmonics of the GRACE/GFO

monthly gravity fields.

Per Knudsen and Ole Andersen

DTU Space, National Space Institute, Geodesy, Kgs. Lyngby, Denmark DTU Space, National Space Institute, Geodesy, Kgs. Lyngby, Denmark

The study is based on the low degree harmonics to focus on the accurate long wavelength parts of the GRACE/GFO gravity fields. To reduce the contamination of mass changes on land a set of point masses are estimated and analyzed to verify estimates of the melting of the major ice sheets. Subsequently, residual gravity fields are used to recover characteristics and regional patterns of changes in the ocean mass.Initially, ocean mass changes are computed using the direct approach using coefficients filtered at harmonic degree and order 20, 40 and 60. The results show that smoothed GSM fields may be used as well, though a wider buffer zone is required. In all cases, values for the global mean ocean mass trends of 2.13-2.47 mm/y were found which are pretty consistent with what others had found already.The second part focus on using the lower degree and more accurate parts of the GRACE/GFO gravity fields. A method for removing land signals contaminating the ocean signal, is tested. It is based on point masses. Though the locations of the point masses at fixed grid nodes may not be optimal, the method seems to work well. The total mass loss on land is about 660 Gt corresponding to 1.83 mm/y ocean sea level gain which is on the low side compared to the values from the initial study. However, using the residual fields from where the continental signals have been removed and no buffer zone, a trend of 1.82 mm/y is obtained. Hence, there is a very good agreement with the corresponding land value. The results show that the method is very successful in removing land signal from the GRACE/GFO monthly fields and facilitates further studies of the changes in the ocean masses.

DOI: 10.5194/gstm2026-16 · https://doi.org/10.5194/gstm2026-16

GSTM2026-17 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

From tight to wide deadband: where current model-based GRACE-FO transplants fall behind

Frederik Jacobs, Miguel Cuadrat-Grzybowski, and Christian Siemes

TU Delft, Aerospace Engineering, Astrodynamics and Space Missions, Delft, Netherlands TU Delft, Aerospace Engineering, Astrodynamics and Space Missions, Delft, Netherlands

One of the GRACE-FO satellites has a malfunctioning accelerometer, rendering it largely unusable for gravity field retrieval. To mitigate the impact, replacement accelerations have been made available, the so-called accelerometer transplants. Several transplants have been developed by different institutes, following either a data-driven approach (the JPL transplant), which assumes that the non-gravitational acceleration is common to both spacecraft and may include empirical corrections, or a model-driven approach (the TU Graz and ZARM transplants), which assumes common atmospheric conditions. Last year, we presented a model-driven transplant developed at TU Delft for the GRACE mission, informed by high-fidelity aerodynamic and radiation-pressure modeling. In this work, we extend that transplant to GRACE-FO.Previous work on GRACE could assess transplant quality by comparing the transplanted acceleration with the measured acceleration for the period when both accelerometers were still operating. In contrast, GRACE-FO offers this opportunity only for a few weeks after launch, when the aerodynamic acceleration is very small, so quality is instead assessed through gravity field retrieval. All transplants, including the JPL, TU Graz and ZARM products, are processed through a single, identical gravity field retrieval using the GROOPS software, so that any differences between solutions can be attributed to the transplant itself. We evaluate the solutions using the open-ocean RMS of equivalent water height, degree-amplitude spectra, and spatial maps after removing bias and trend from a reference field.To mitigate small fuel leaks, the GRACE-FO satellites changed attitude mode in 2023. Before 2023, the satellites were in tight-deadband mode, requiring many thruster firings but keeping a stable attitude. Since 2023, they have been in wide-deadband mode, needing fewer firings but allowing larger attitude variations. We compare transplant quality across both modes using six months distributed across the mission, selected to separate the effect of the attitude mode from that of solar activity and sun-orbit geometry.In tight-deadband mode, the TU Delft model-driven transplant is competitive with the operational data-driven JPL product. In wide-deadband mode, we observe increased noise in the gravity field models from the TU Delft transplant

compared to those obtained with the JPL transplant. We attribute this observation to the amplification of modeling errors from different sources, such as gas-surface interaction models, atmospheric models, and attitude errors, to which model-based transplants are inherently sensitive.

DOI: 10.5194/gstm2026-17 · https://doi.org/10.5194/gstm2026-17

GSTM2026-18 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Advancing glacier and ice-sheet mass-change observations with GRACE-C, NGGM and MAGIC

Ingo Sasgen1 and Bert Wouters2

Alfred-Wegener-Institute Bremerhaven, Geosciences Department / Section Glaciology, Bremerhaven, Germany · Department of Geoscience and Remote Sensing, Delft University of Technology, Delft, The Netherlands

GRACE and GRACE-FO have transformed observations of glacier and ice-sheet mass change, but their monthly solutions and correlated noise limit the detection of short-lived melt and accumulation events, and long-term ice-dynamic signals. We assess ESA's inclined Next Generation Gravity Mission (NGGM; ~70° inclination), which provides denser, more uniform sampling throughout the non-polar latitude band than a near-polar pair. End-to-end simulations with ESA-funded SING project show that NGGM improves the recovery of lower-latitude glacier mass change, more clearly resolving seasonal variability and long-term trends, showing additional potential for hydrological applications. For Greenland and Antarctica, NGGM improves sensitivity to short-lived melt and accumulation events and provides independent constraints on long-term accelerations in ice-dynamic discharge. Within MAGIC, NGGM complements the near-polar GRACE-C pair to provide more homogeneous near-global observations of glaciers and ice sheets, strengthening constraints on regional climate and ice-sheet models and future sea-level rise.

DOI: 10.5194/gstm2026-18 · https://doi.org/10.5194/gstm2026-18

GSTM2026-19 B.2 · CRYOSPHERE · ORAL

Arctic glacier mass change from GRACE/GRACE-FO: Regional variability, atmospheric drivers and rapid GIA

Ingo Sasgen

Alfred-Wegener-Institute Bremerhaven, Geosciences Department / Section Glaciology, Bremerhaven, Germany Alfred-Wegener-Institute Bremerhaven, Geosciences Department / Section Glaciology, Bremerhaven, Germany

Arctic glaciers are losing mass rapidly, but their regional variability reflects both atmospheric forcing and solid-Earth processes. We combine GRACE/GRACE-FO glacier mass balances with ESA CCI Permafrost and CALM observations to investigate common drivers of glacier melt and permafrost thaw. Two dominant modes of summer atmospheric circulation explain about 75% of their common pan-Arctic variability over 2002–2022. We further show that rapid viscoelastic rebound in low-viscosity regions such as Alaska and Iceland can substantially bias gravimetry-derived glacier trends if not accounted for. Together with recent GlaMBIE assessments, these results provide a broader view of Arctic glacier change, its climatic drivers, and remaining observational uncertainties.

DOI: 10.5194/gstm2026-19 · https://doi.org/10.5194/gstm2026-19

GSTM2026-20 B.4 · HYDROLOGY · ORAL

Using GRACE-/FO 10-day CNES solutions for hydrological applications

Judith Garnier1, Alejandro Blazquez3, Alexandre Boughanemi1, Fabien Gachet4, Emeric Lavergne2, Julien Lefebve3, Jean-Michel Lemoine4, Delphine Leroux2, Eric Pellereau1, and Julia Pfeffer1

Magellium, Toulouse, France · CNES, Toulouse, France · Université de Toulouse, LEGOS (CNES/CNRS/IRD/UT), Toulouse, France · Université

de Toulouse, GET (CNES/CNRS/IRD/UT), Toulouse, France

GRACE and GRACE-FO satellite missions measure the spatial and temporal variations in the Earth's gravity field at a spatial resolution of around 300 kilometers. Unlike most gravity field products available at monthly resolutions, the CNES produces a 10-day Level 2 solution constrained by Satellite Laser Ranging using Singular Value Decomposition and Tikhonov regularization, improving the estimation of the low- and high-degree spherical harmonics. Resolved up to degree 90 without spatial filtering, this product enhances the spatial resolution of satellite gravity products. Here, we present the hydrological applications of the Level 3 product, corrected for the missing degree 1 and Glacial Isostatic Adjustment (GIA), representing Terrestrial Water Storage Anomalies (TWSA) expressed in gridded Equivalent Water Heights (mmEWH).This study assesses the added value of the 10-day CNES product for monitoring extreme hydrological events such as droughts and floods based on GRACE-Drought Severity Index (GRACE-DSI). Results show that the 10-day product captures major events like the 2010-2011 La Niña in Australia, and the severe drought that affected France during the 2022 summer. Comparison with the monthly CNES ensemble and daily ITSG-2018 products confirms the strong temporal consistency at seasonal to interannual time scales and substantially higher spatial and temporal resolutions of the CNES 10-day product.GRACE-DSI also shows good agreement with in situ indicators, particularly those integrating information over long time scales, such as the Palmer Drought Severity Index and the Standardized Precipitation Evapotranspiration Index (SPEI). By accounting for the total terrestrial water storage variability, integrated from the surface to deep aquifers, the GRACE-DSI complements indices targeting individual components of the hydrological cycle. The improved temporal and spatial resolution makes the 10-day CNES product and the associated 10-day GRACE-DSI particularly relevant for the monitoring of extreme hydrological events. This study also suggests potential for the preparation of hydrological applications of the future satellite gravimetry missions NGGM and MAGIC, with expected higher spatio-temporal resolution and shorter latency time.

DOI: 10.5194/gstm2026-20 · https://doi.org/10.5194/gstm2026-20

GSTM2026-21 B.5 · MULTIDISCIPLINARY SCIENCE · ORAL

Developing the SAGSA L3 gravimetry ensemble and its applications

Alexandre Boughanemi1, Alejandro Blazquez2, Marie Bouih1, Alexandre Couhert3, Fabien Gachet3, Judith Garnier1, Louis-Marie Gauer2, Hugo Lecomte2,4, Jean-Michel Lemoine3, Eric Pellereau1, Yann Ziegler2, and Julia Pfeffer1

MAGELLIUM, Ramonville Saint-Agne, 31520, France · Université de Toulouse, LEGOS (CNES/CNRS/IRD/UT), Toulouse, France · Université de Toulouse, GET (CNES/CNRS/IRD/UT), Toulouse, France · Finnish Geospatial Research Institute, National Land Survey of Finland

Since 2002, the GRACE and GRACE-FO satellite missions have been monitoring monthly variations in the gravitational field caused by mass redistributions within the Earth's system. To extract information on the global and regional water cycle, estimates of the gravity field must be corrected to account for several instrumental and geophysical processes. The SAGSA L3 dataset uses an ensemble approach to estimate water mass redistributions in the ocean, atmosphere, hydrosphere, and cryosphere, as well as their uncertainties. Here, we review the processing and post-processing steps included in the ensemble approach, with a particular focus on the dry mass of the atmosphere and leakage from the continents to the oceans. A small-amplitude drift has been identified in the global average estimate of atmospheric mass, affecting the standard ERA-5 reanalysis and all dealiasing models derived from it. We show that the global atmospheric mass correction has a significant impact on ocean applications, reducing the sea level budget residuals by approximately 2 mm between 2016 and 2026. In addition, a new framework has been established to better correct leakage errors, which are linked to the mislocation of water mass variations along coastlines. The new leakage correction allows for a better redistribution of water masses along the coastline, which has a significant impact on estimates of glacier mass balance (up to 30 Gt/yr across all glaciers) as well as on sea level variations near the coast (up to 2 mmSLE) between 2002 and 2026. While current efforts enable the SAGSA ensemble to contribute to leading international research initiatives—such as the validation of simulations used for future missions or glacier mass balance comparison exercises— additional efforts are needed to improve the accuracy of water mass distribution estimates, particularly with regard to post-processing corrections related to solid

Earth deformations (i.e., glacial isostatic adjustment, co- and post-seismic deformations, as well as hydrological and oceanic loads).

DOI: 10.5194/gstm2026-21 · https://doi.org/10.5194/gstm2026-21

GSTM2026-22 A.4 · THE MEAN EARTH GRAVITY FIELD · ORAL

The satellite-only gravity field model GOCO2025s

Felix Öhlinger1, Torsten Mayer-Gürr1, Sandro Krauss1, Patrick Dumitraschkewitz1, Andreas Strasser1, Barbara Süsser-Rechberger2, Jan Martin Brockmann3,4, and Josef Niedermaier5

Institute of Geodesy, Graz University of Technology, Graz, Austria · Institute of Physics, University of Graz, Graz, Austria · Institute of Geodesy and Geoinformation, University of Bonn, Bonn, Germany · High Performance Computing and Analytics Lab, University of Bonn, Bonn, Germany · Institute for Astronomical and Physical Geodesy, Technical University of Munich, Munich, Germany

GOCO2025s is a global gravity field model represented by spherical harmonic coefficients up to degree and order 300, with constrained secular, annual, and semi-annual variations up to degree and order 200. It is derived from 23 years of satellite data, incorporating observations from the dedicated gravity field missions Gravity field and steady-state Ocean Circulation Explorer (GOCE), Gravity Recovery and Climate Experiment (GRACE), and GRACE Follow-On (GRACE-FO), as well as kinematic orbits of 18 low Earth orbit satellites and satellite laser ranging observations to 9 passive geodetic satellites. The complementary measurement principles of the different satellite missions and their strengths in determining specific parts of the spherical harmonic spectrum were exploited by combining the data at the normal equation level, with emphasis on the stochastic modeling of each contribution. Particular focus was placed on the inclusion of GRACE-FO laser ranging interferometer (LRI) data, which provided the dominant contribution to the estimation of the spherical harmonic coefficients between degrees 70 and 200. All processing steps, except for the determination of the GOCE normal equations, were carried out at the Institute of Geodesy at Graz University of Technology (TUG) employing up-to-date background models. The consistent in-house processing of nearly all satellite data, using the Gravity Recovery Object Oriented Programming System (GROOPS) software package, minimized errors arising from inconsistencies of different algorithms or models being applied.The individual files of the GOCO2025s model (Öhlinger et al., 2025) are available at the TUG repository and identified with the following DOI: 10.3217/f48p8-8h651. They comprise the static and time-dependent gravity field coefficients and their respective uncertainty information, all parts of the normal equation system of the static gravity field, and the gravitational potential of non-tidal atmospheric and ocean masses already included in GOCO2025s.

DOI: 10.5194/gstm2026-22 · https://doi.org/10.5194/gstm2026-22

GSTM2026-23 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Impact of the new GFZ RL07 GRACE/GRACE-FO processing standards on the Level-2 gravity field solutions

Markus Hauk1, Christoph Dahle1, Michael Murböck1,2, Natalia Panafidina1, Josefine Wilms1, Karl-Hans Neumayer1, and Annette Eicker1

GFZ Helmholtz Centre for Geosciences, Potsdam, Germany · Technische Universität Berlin, Berlin, Germany

GFZ aims to distribute a reprocessed and improved RL07 Level-2 gravity field time series in the last quarter of 2026. Most of these improvements have been developed within the Research Unit “New Refined Observations of Climate Change from Spaceborne Gravity Missions” (NEROGRAV) funded by the German Research Foundation DFG.When developing the new GFZ RL07 processing standards, the main focus was on an optimized stochastic modeling during the GRACE/GRACE-FO gravity field determination. This includes the extension of the stochastic instrument error models, the optimization of the combination of the different observations in terms of relative weighting, and the inclusion of non-tidal background model error variance-covariance information in the adjustment process. Further changes compared to GFZ RL06 comprise the parameterization as well as the applied background models, e.g., the ocean tide model and the static gravity field model.This presentation provides an overview of the GFZ RL07

performance compared to RL06 with focus at specific processing aspects and its impact on the gravity field solution, such as the ocean tide model update or the influence of stochastic modelling of observation and background model errors. Improvements stemming from the applied advanced processing strategy led to a significant reduction of noise (> 30% relative to GFZ RL06), as well as more realistic formal uncertainties of the estimated gravity field parameters. In addition, a short outlook on the expected performance of LRI-based GRACE-FO gravity field performance is presented.

DOI: 10.5194/gstm2026-23 · https://doi.org/10.5194/gstm2026-23

GSTM2026-24 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Future Satellite Gravimetry for Global Total Drainable Water Storage Estimation: Mission Performance and Basin Controls

Alireza Sobouti1, Mohammad J. Tourian1, Laura Jensen2, Julia Pfeffer3, Peyman Saemian1, Henryk Dobslaw2, Ilias Daras4, and Nico Sneeuw1

University of Stuttgart, Institute of Geodesy, Stuttgart, Germany · GFZ Helmholtz Centre for Geosciences, Telegrafenberg, Potsdam, 14473, Germany · Magellium, 1 rue Ariane, 31520, Ramonville-saint-Agne, France · ESA/ESTEC, Noordwijk, The Netherlands

Total Drainable Water Storage (TDWS) represents the portion of basin water storage that can drain and sustain river discharge, providing a basin-scale indicator of freshwater availability. Reliable global estimation of TDWS could therefore improve assessments of freshwater availability and help identify basins with limited capacity to sustain discharge. However, estimating TDWS globally from space-based Terrestrial Water Storage Anomalies (TWSA) remains constrained by both satellite-observation errors and differences in basin hydrological behavior. We use closed-loop satellite gravimetry simulations incorporating realistic instrumental and aliasing errors to estimate TDWS for mission configurations representative of the Gravity Recovery and Climate Experiment Continuity (GRACE-C), the Next Generation Gravity Mission (NGGM), and the Mass-change and Geosciences International Constellation (MAGIC). Assuming a linear runoff–storage (R–S) relationship, TDWS is estimated for more than 200 globally distributed river basins. Two independent Earth system model realizations, ESA ESM2.0 and ESM3.0, are used as reference TWSA datasets. Mission-based TDWS estimates are evaluated against their corresponding references, and retrieval errors are examined in relation to basin area, directional shape, storage and discharge variability, terrain relief, land cover, climate, and R–S linearity. NGGM and MAGIC consistently improve TDWS estimation relative to GRACE-C. Median relative errors decrease from 7.9% for GRACE-C to 4.2% and 4.3% for NGGM and MAGIC, respectively, under ESM2.0, and to 4.5% and 4.1% under ESM3.0. The largest size-dependent improvements occur for basins between 200,000 and 500,000 km², with error reductions of about five percentage points relative to GRACE-C. Directional basin geometry also affects retrieval performance, with improvements increasing toward strongly north–south-oriented basins under ESM3.0. Environmental controls remain evident: rugged basins generally show larger TDWS errors than flat basins, while under ESM3.0 MAGIC reduces the median error in polar climates from 10.0% to 4.4%. The reference R–S relationship achieves a Nash–Sutcliffe Efficiency (NSE) of at least 0.7 in 86% of basins under ESM2.0 and 79% under ESM3.0. NGGM and MAGIC reproduce the reference NSE more closely than GRACE-C, demonstrating the benefit of improved gravity-field recovery. Nevertheless, improved satellite observations cannot compensate for a weak or unsuitable R–S relationship. These simulations therefore provide more than a comparison of future mission performance: they identify the observational, geometric, and hydrological conditions that control basin-scale TDWS retrieval. Moving toward robust global TDWS estimation will require both next-generation satellite gravimetry and basin-specific assessments of hydrological retrieval suitability.

DOI: 10.5194/gstm2026-24 · https://doi.org/10.5194/gstm2026-24

GSTM2026-25 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

COST-G: new options and motivation for gravity field combinations on normal

equation level

Ulrich Meyer1, Martin Lasser1, Christoph Dahle2, Franck Reinquin3, Torsten Mayer-Gürr4, and Adrian Jäggi1

University of Bern, Astronomical Institute, Bern, Switzerland · German Research Centre for Geosciences, Potsdam, Germany · Observatoire Midi-Pyrénées, Toulouse, France · Graz University of Technology, Austria

So far, the operational combination of the monthly GRACE, or GRACE-FO gravity field solutions by the Combination Service of Time-variable Gravity fields (COST-G) has been performed on the solution level. The originally foreseen combination on the normal equation (NEQ) level suffered from the very diverse and not realistic formal uncertainties, which hampered the application of Variance Component Estimation (VCE) and made the use of empirical weighting factors mandatory. Meanwhile, empirical noise models that enable the determination of realistic formal uncertainties are available from several GRACE-FO analysis centers. Experimental NEQ-combinations confirmed the applicability of VCE for the GRACE-FO combination and also the combination with SLR turned out to be feasible without the use of empirical weighting factors. The combination of the future GRACE-C and NGGM missions on the NEQ-level is a prerequisite to generate gravity field solutions of the MAGIC constellation. Consequently, the combination of NEQs from different Analysis Centers (ACs), including also SLR, will be the goal of future COST-G releases, but relies on the cooperation of the ACs to provide their NEQs and to apply realistic noise models.

DOI: 10.5194/gstm2026-25 · https://doi.org/10.5194/gstm2026-25

GSTM2026-26 B.5 · MULTIDISCIPLINARY SCIENCE · ORAL

Unified Recovery of Low-Degree Time-Variable Gravity and Terrestrial Reference Frame Parameters Through Multi-Technique Space Geodesy

Athina Peidou, Willy Bertiger, Shailen Desai, Bruce Haines, Mike Heflin, and Paul Ries

Jet Propulsion Laboratory, Pasadena, United States of America Jet Propulsion Laboratory, Pasadena, United States of America

This study processes over a decade of space geodetic observations from Low Earth Orbiters (LEO; e.g., GRACE, Jason, and Sentinel series), dedicated Satellite Laser Ranging platforms (SLR; e.g., LAGEOS-1/2, LARES), and GPS ground and satellite tracking networks to simultaneously realize the terrestrial reference frame and recover low-degree time-variable gravity coefficients at the observation level. Building upon this unified multi-technique processing framework, we expand our processing strategy to incorporate GRACE-FO onboard accelerometer data and inter-satellite ranging data. By leveraging the dynamic connection between frame realization and gravity estimation inside a single, consistent processing framework, our approach directly resolves low-degree spherical harmonic terms, including geocenter motion (degree-1), J2, and J3. Finally, we evaluate the consistency of these observation-level low-degree gravity solutions against established products, specifically Technical Notes 13 and 14 as well as independent SLR-derived estimates, demonstrating the capability of this integrated framework to provide a robust, independent alternative for low degree gravity field determination.© 2026. California Institute of Technology. Government sponsorship acknowledged.

DOI: 10.5194/gstm2026-26 · https://doi.org/10.5194/gstm2026-26

GSTM2026-27 B.4 · HYDROLOGY · POSTER

Insights from the DANSk-LSM Project to Advance GRACE(-FO) and Multi-Sensor Data Assimilation for Hydrological Monitoring and Early Warning Systems

Ehsan Forootan, Fan Yang, Leire Retegui Schiettekatte, and Maike Schumacher

Aalborg University, Geodesy Group, Department of Sustainability and Planning, Aalborg, Denmark Aalborg University, Geodesy Group, Department of Sustainability and Planning, Aalborg, Denmark

The DANSk-LSM (Developing Efficient Multi-Sensor Data Assimilation Frameworks for Integrating Earth Observation Satellite Data into Land Surface Models) project, funded by Independent Research Fund Denmark (DFF), developed and demonstrated physically consistent and computationally efficient data assimilation frameworks for hydrological modelling. Between 2022 and 2026, the project integrated multi-source Earth observation data into land surface models to improve monitoring and forecasting of hydrological extremes.A major outcome was the assimilation of GRACE/GRACE-FO terrestrial water storage (TWS) observations together with satellite-derived surface soil moisture and recently the addition of surface water dynamics is being explored. Novel data assimilation settings were developed and evaluated across diverse river basins (e.g., Brahmaputra, Danube and Mississippi), improving the representation of water storage variations, fluxes, and drought and flood evolution. Scientific and implementation challenges were also identified.DANSk-LSM delivered open-access tools, including PyGLDA software, and high-resolution signle sensor and multi-sensor data assimilation tools. The project also fostered collaboration across geodesy, remote sensing, hydrology, and data science communities, laying the groundwork for next-generation Digital Twin Earth hydrology systems and future satellite gravity missions. This poster summarizes the main scientific advances and lessons learned, highlighting the role of gravity-derived water storage observations in (operational) hydrological monitoring and disaster preparedness.Keywords: GRACE and GRACE-FO, terrestrial water storage, multi-sensor data assimilation, land surface models, hydrological extremes, drought monitoring, flood forecasting, Earth Observation, Digital Twin Earth.

DOI: 10.5194/gstm2026-27 · https://doi.org/10.5194/gstm2026-27

GSTM2026-28 A.1 · GRACE-FO PROJECT STATUS · ORAL

JPL GRACE and GRACE-FO Level-2 Overview

David Wiese, Dah-Ning Yuan, Christopher McCullough, Eugene Fahnestock, Athina Peidou, and Felix Landerer

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States of America Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States of America

This talk will provide a status update on Level-2 data processing at JPL. Included will be an overview of data products currently delivered to the community, focusing on the operational RL06.3 GRACE-FO fields. Assessments of data quality and error levels, along with detailed discussions of the solution strategy will be included. A look towards an RL07 reprocessing of GRACE and GRACE-FO will additionally be provided. The reprocessing incorporates the estimation of a full noise covariance model of the satellite observations in the gravity estimates along with updated background force models.

DOI: 10.5194/gstm2026-28 · https://doi.org/10.5194/gstm2026-28

GSTM2026-29 B.5 · MULTIDISCIPLINARY SCIENCE · ORAL

GRACE Data as Non-Tidal Deformation in Reference Frame Determination:

From Position Timeseries Forecast to Frame Realization

Toshio Mike Chin, Claudio Abbondanza, Richard Gross, and Michael Heflin

NASA Jet Propulsion Laboratory, California Institute of Technology NASA Jet Propulsion Laboratory, California Institute of Technology

The terrestrial reference frame (TRF) is the foundation for all space-based, airborne, and ground-based Earth observations. The accuracy of position data ultimately depends on the accuracy of TRF, which consists of precise Earth-centered coordinates of hundred of surface sites, many of which anchor the orbit estimates of the global navigation satellites.The non-tidal deformation (NTD) due to loading from atmosphere, ocean, and ground water tends to seasonally modulate the space geodetic position data used in determination of TRF. Campaign to determine the International Terrestrial Reference Frame (ITRF) takes place approximately every 5 years (with a recent initiative to realize intra-campaign ITRF updates every year). Because of such latency in ITRF realizations, forecasts of the TRF positions are desired in a typical application, relying on the TRF velocity estimate to forecast the linear trajectory projection. Here, we examine how various models of NTD could affect the accuracy of such TRF position trajectory forecasts.In particular, we have seen that GRACE-based surface deformation data can be used as proxy of NTD in estimation of the site velocity. At individual GNSS sites, use of the GRACE data were particularly effective in mitigating extremely large forecast errors at certain sites, out-performing the standard ITRF NTD estimation method (via a fit to a truncated Fourier series) by an average of 3 mm. Also, no degradation of the forecasts due to addition of the GRACE data was observed. In order to extend the use of the GRACE-based deformation data to a TRF realization, we need to address: (1) the temporal coverage of the GRACE data, such as the gap between GRACE and GRACE-FO and pre-2003 period; (2) compatibility in site-to-site spatial correlations exhibited by GRACE versus TRF site position data. We show that the spatio-temporal behavior of the GRACE deformation data can be "learned" empirically via multichannel singular spectrum analysis (MSSA) to fill the GRACE/GRACE-FO gap. The MSSA GRACE data have improved the forecast accuracy in the gap where we have resorted to linear interpolation in the past, and the improved forecasts in the gap period were statistically indistinguishable from the period with actual GRACE data. This encourages consideration for hindcasting the GRACE data into a much longer 1997-2003 period for the use in a full TRF realization.

DOI: 10.5194/gstm2026-29 · https://doi.org/10.5194/gstm2026-29

GSTM2026-30 B.2 · CRYOSPHERE · ORAL

Leveraging Satellite Altimetry and Gravimetry for High-Resolution Antarctic Mass Change

David Wiese1, Johan Nilsson2,1, Nicole-Jeanne Schlegel3, Alex Gardner1, Matthias Ellmer1,4, and Felix Landerer1

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States of America · Uppsala University, Sweden · NOAA/OAR Geophysical Fluid Dynamics Laboratory · now at: Planet Labs PBC

Satellite altimetry missions have transformed our capability to observe ice sheet surface elevation change with unprecedented spatial precision. While altimetry excels at localized surface height changes, converting height to mass requires firn air content corrections. Conversely, satellite gravimetry (GRACE/GRACE-FO) directly measures mass change with high accuracy over large scales, but is limited to coarse spatial resolutions (~300 km).Here, we present a formal data combination framework integrating two decades (2002–2020) of satellite altimetry observations (ICESat, ICESat-2, CryoSat-2, Envisat, ERS-2) with GRACE/GRACE-FO Level-1 data at the level of the gravimetry normal equations. By solving for 1 degree circular disk mascon elements, altimetry observations dominate the inversion at individual mascon scales, breaking inter-mascon spatial correlations and localizing signals over dynamic glacier catchments like Pine Island, Thwaites, and Totten Glaciers. Gravimetry dominates at long spatial wavelengths, preserving continental and basin-scale mass balance accuracy.The monthly Antarctic mass change solution achieves an effective spatial resolution of (100 km)2 , fulfilling long-standing community target requirements. Applying mass-conserving gain factors derived from altimetry spatial fields further yields downscaled monthly mass change at a 1.92 km scale. Rigorous propagation of stochastic measurement errors and systematic model uncertainties (GIA, firn air content) is included.This joint inversion highlights the value of multi-sensor Earth data fusion, offering a data

processing blueprint to ingest observations from future satellite altimetry missions CRISTAL and EDGE, and satellite gravimetry missions GRACE-C and NGGM.

DOI: 10.5194/gstm2026-30 · https://doi.org/10.5194/gstm2026-30

GSTM2026-31 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · POSTER

Interaction of Deterministic Errors in GRACE-Continuity

David Wiese1, Andrew Ludwig1,2, Chirag Patel3, Christopher McCullough1, Nadege Pie3, and Himanshu Save3

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States of America · now at: Zoox, Inc. (An Amazon Subsidiary) · Center for Space Research, The University of Texas at Austin

The overall error in gravity field products derived from satellite-to-satellite tracking gravity missions like the Gravity Recovery and Climate Experiment Follow-On (GRACE-FO) is driven by a combination of temporal aliasing error, and measurement system error consisting of both stochastic and deterministic sources. While the impact of each of these error sources has been extensively documented in the literature, there has been relatively little study of how the phasing of deterministic errors, or tones, propagates into the gravity field solution. Because these errors are primarily driven by on-orbit thermal variations, this study uses flight data from GRACE-FO thermistors alongside high-fidelity thermo-mechanical simulations that map temperature variations to structural deformation to investigate the phase and amplitude of these tones.We utilize these data to derive the most probable phasing of twice-per-revolution tones in both inter-satellite ranging and accelerometer measurements. We are particularly interested in the interplay of these tones and the twice-per-revolution component of absolute frequency error in the inter-satellite ranging signal. Unlike once-per-revolution errors, which are largely filtered out by empirical parameterization during standard orbit determination, twice-per-revolution errors survive the filtering process and map directly into the estimated gravity field. While we exclusively use GRACE-FO data and numerical simulations of the GRACE-Continuity flight system in this work, the principles of this work apply to any gravity mission using an inter-satellite ranging system. We demonstrate that tone phasing plays a critical role in the magnitude of the resulting gravity field degree variance, particularly in C20. We conclude by discussing strategies to account for deterministic measurement system errors.

DOI: 10.5194/gstm2026-31 · https://doi.org/10.5194/gstm2026-31

GSTM2026-32 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Ellipsoidal Equal-Area Panel Mascons from GRACE Intersatellite Geopotential Differences: A Preliminary Assessment over China

Vagner Ferreira1, Yongzhao Fan1, Thomas Grombein2, Kurt Seitz2, and Hansjörg Kutterer2

Hohai University, School of Earth Sciences and Engineering, Nanjing, China · Karlsruhe Institute of Technology (KIT), Geodetic Institute, Karlsruhe, Germany

We present a mascon-type solution that partitions the Earth's surface into contiguous, gap-free, equal-area panels defined directly on the reference ellipsoid via an authalic-latitude mapping, eliminating the need for a constant-radius approximation or subsequent rescaling. Each panel is treated as a locally curved point mass, yielding a closed-form kernel for the GRACE intersatellite geopotential differences (GPD), which serve as observations. Panel masses are estimated in a Bayesian (maximum a posteriori) framework using a diagonal prior covariance informed by the temporal variability of an independent hydrological field. The regularized normal equations are solved by preconditioned conjugate-gradient least squares (CGLS) with L-curve stopping, avoiding the assembly or factorization of any large matrix and enabling the full monthly inversion to run on standard workstation hardware. Based on this framework, a preliminary global inversion was performed for 4,704 ellipsoidal panels for each of 155 monthly WHU/SGG GPD datasets spanning April 2002 to July 2016. For a preliminary assessment, the results were clipped to China and evaluated against the official JPL, CSR, and GSFC mascon solutions. Overall, the solution reproduces the dominant spatial patterns, seasonal amplitude, and secular trends of the reference products, with spatial correlations ranging from 0.73 to 0.82 for annual amplitude and 0.67 to 0.74 for trend, and a spatial

correlation of 0.80 with JPL-M for a single-month (September 2005) verification. Basin-averaged time series for China's nine major river basins reproduce reported drought events, with discrepancies against the reference mascons confined mainly to an arid basin. These results demonstrate that the proposed ellipsoidal-panel parametrization offers an efficient, transparent, and reproducible alternative for regional mass inversion.

DOI: 10.5194/gstm2026-32 · https://doi.org/10.5194/gstm2026-32

GSTM2026-33 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Data-driven Uncertainty-aware 5-day Surface Mass Variations from GRACE Ranging Residuals

Michal Cuadrat-Grzybowski, Frederik Jacobs, and João G. de Teixeira Da Encarnação

Delft University of Technology, Space Engineering, Delft, Netherlands Delft University of Technology, Space Engineering, Delft, Netherlands

This study directly utilises GRACE Level-1B (L1B) KBR post-fit residuals, provided by the Center for Space Research, as the basis for high-frequency mass change modelling. We showcase the updated methodology and its uncertainty characteristics, demonstrating the potential of KBR post-fit residuals to recover sub-monthly surface mass variations from the GRACE(-FO) mission by exploiting information already contained in the conventional monthly gravity-field processing chain, without requiring additional observations or a separate high-frequency gravity-field inversion. We submitted the work to the Journal of Geodesy, and it continues the methodology presented in our previous Science Meeting contribution.

The methodology combines geo-fits with post-fit residuals from the KBR instrument of the GRACE satellites. Geo-fits represent the range-rate L1B observations fitted to monthly spherical harmonic gravity models in traditional least-squares gravity retrieval. The post-fit residuals represent the instantaneous discrepancies between the reduced observations (pre-fits) and the corresponding geo-fits, and therefore contain information not represented by the monthly gravity-field solution. We spatially interpolate 5-day sets of post-fit residuals and combine them with the corresponding spatially binned monthly geo-fits, resulting in gridded high-frequency (HF) geo-fits. We then transform the HF geo-fits into equivalent water height (EWH) using an empirical relationship.

Since the previous study, the methodology has been further developed in several aspects. The updated framework considers residuals in both the range and line-of-sight gravity difference (LGD) domains, providing three formulations: Range, LGD, and a hybrid formulation. The latter uses a weighted and normalised combination of both observables, with the weighting selected to minimise the regression error in the transformation to EWH. The observation-to-EWH transformation has also been improved, showing that the relationship is not only spatially variable but also time-dependent, reflecting changes in inter-satellite range and satellite altitude throughout the GRACE mission. These variations affect KBR sensitivity to mass change but should not appear as EWH variations; the updated transformation therefore accounts for these effects. The three formulations (range, LGD and hybrid) are available as 5-day EWH products (https://doi.org/10.4121/4fc748e8-01c7-4f06-87da-653937b078f7).

Further developments include uncertainty propagation, which provides spatially and temporally resolved uncertainty estimates for the 5-day EWH fields. The additional uncertainty associated with reducing the temporal resolution from monthly to 5-day estimates remains moderate. Relative to the corresponding monthly solutions, the uncertainty increases by approximately 13–20% over land and 8–12% over the oceans. For comparison, reducing the averaging period from one month to five days while assuming independent observations and no additional information would yield an expected uncertainty increase of approximately 145%. The relatively small increase results from exploiting the spatial and temporal information contained in the KBR post-fit residuals rather than treating each 5-day solution as an independent gravity-field inversion.

The approach does not use hydrological models or other external geophysical priors, nor does it require temporal regularisation or Kalman filtering. It therefore avoids the need for a separate high-frequency gravity inversion framework. The resulting 5-day EWH grids are evaluated against daily solutions and through case studies of the 2007 Bangladesh floods and the January 2013 Cyclone Oswald.

DOI: 10.5194/gstm2026-33 · https://doi.org/10.5194/gstm2026-33

GSTM2026-34 B.4 · HYDROLOGY · ORAL

Downscaling and Gap-Filling GRACE-Based Terrestrial Water Storage Anomalies Using Bias-Aware Machine Learning

Tatiana Solovey1, Vytautas Samalavicius2, Justyna Śliwińska-Bronowicz3, Rafał Janica4, Agnieszka Brzezińska5, and Anna Stradczuk6

Polish Geological Institute - National Research Institute, Hydrogeology, Warsaw, Poland · Institute of Geosciences, Faculty of Chemistry and Geosciences, Vilnius University · Centrum Badań Kosmicznych Polskiej Akademii Nauk · Polish Geological Institute - National Research Institute, Hydrogeology, Warsaw, Poland · Polish Geological Institute - National Research Institute, Hydrogeology, Warsaw, Poland · Polish Geological Institute - National Research Institute, Hydrogeology, Warsaw, Poland

This study presents a methodological framework to temporally reconstruct and spatially downscale GRACE Terrestrial Water Storage (TWS) anomalies using a bias-aware Random Forest model for the transboundary Bug River Basin (Poland–Ukraine–Belarus). The approach incorporates three main innovations that enhance satellite-based TWS estimation. First, missing monthly TWS anomalies were reconstructed using an RF model driven solely by lagged GRACE values and calendar-month information, which avoided information leakage from auxiliary hydrological predictors and yielded a temporally continuous TWS record despite data gaps, including the 2017–2018 GRACE–GRACE-FO hiatus. Second, RF was used to downscale the reconstructed GRACE-derived TWS to a 0.1°×0.1° resolution, using precipitation, evapotranspiration, river runoff, and soil moisture as predictors, together with elevation, lithology, and land-cover class as static physiographic attributes. Third, a bias-aware, mass-conserving block-wise correction was applied to reconcile the downscaled fields with the original GRACE water mass at coarse resolution, by redistributing monthly systematic errors within spatial blocks using river-runoff-based weights, thereby enforcing exact mass-balance closure while preserving physically meaningful sub-grid variability.The results indicate that GRACE-derived TWS dynamics are best captured by a predictor with an integrated origin — river runoff — rather than by individual hydrometeorological drivers, underscoring the importance of multivariate approaches to downscaling. The combined use of hydrometeorological parameters allows the model to capture complementary aspects of the storage balance that are not evident from any single predictor alone. In terms of predictive skill, the 1-month lag dominates, consistent with strong month-to-month persistence in TWS. Month (the seasonal indicator) shows moderate importance. Among the independent predictors, hydrological variables were the dominant controls on the predicted GRACE-derived TWS values, with the three-month rolling mean of river runoff being the most influential predictor. The northern regions of the Bug River Basin exhibit stronger and more regular seasonal cycles, which enhances the effectiveness of the lagged predictors and seasonal encoding used in the imputation model. By contrast, the southern areas tend to show more episodic hydrological variability, potentially driven by localized precipitation extremes, rapid runoff responses, or human water management, which reduces temporal autocorrelation and limits predictability based solely on past GRACE observations. The spatial distribution of linear trends in GRACE-derived TWS reveals a coherent pattern of long-term storage decline over the full 2002–2024 period, with pronounced spatial heterogeneity: the magnitude of decline increases toward the southern part of the study area, where the strongest negative trends are observed.The proposed methodological framework, strengthened by rigorous spatiotemporal cross-validation and physically based mass-conservation constraints, enables improved monitoring and forecasting of groundwater in transboundary catchments and facilitates the reconstruction of continuous TWS changes over time and space, which is particularly valuable for regions with sparse or conflict-disrupted in-situ observation networks.

DOI: 10.5194/gstm2026-34 · https://doi.org/10.5194/gstm2026-34

GSTM2026-35 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

MicroSTAR: NGGM Space Accelerometer Last Developments

Kristen Maquaire, Damien Boulanger, Ratana Chhun, Bruno Christophe, Marine Dalin, Théophile Hervieux, Vincent Lebat, Françoise Liorzou, Grégoire Pique, Manuel Rodrigues, and Milène Rodriguez-Beignet

Department of Physics, Instrumentation, Environment and Space, ONERA the French Aerospace Lab, Paris Saclay University, F-92322 Châtillon, France Department of Physics, Instrumentation, Environment and Space, ONERA the French Aerospace Lab, Paris Saclay University, F-92322 Châtillon, France

ONERA laboratory have been providing highly sensitive electrostatic accelerometers involved in all major space Geodesy missions for more than five decades. Since the first mission launched, the accuracy of the accelerometers has significantly improved over time. For instance, the ESA GOCE mission accelerometers exhibited a few 10-12ms-2Hz-1/2 resolution. In a Fundamental Physics application, the T-SAGE accelerometers developed at ONERA allowed to test the Equivalence Principle on board the MICROSCOPE satellite at a level of 10-15, establishing in 2022 the best accuracy ever achieved for the test, besting the previous one attained with Lunar Laser Ranging by two orders of magnitude.Currently, MicroSTAR accelerometers are selected for ESA Next Generation Gravity Mission (NGGM). This mission aims to extend observations provided by GRACE-family satellites, with the same strategy. ONERA's project is ongoing and the instrument phase C started early 2026. Currently, the mechanical sensor reached TRL-6 while the electronic unit and the overall system achieved TRL-5 with a performance level foreseen of about a few 10-12ms-2Hz-1/2. In the same fashion as the past instruments, the accelerometer is based on the electrostatic suspension of a proof-mass. The position and the attitude of the mass are measured and controlled with capacitive sensors and actuators. One of the main improvements of MicroSTAR is its cubic proof-mass, as opposed to the previous parallelepipedal ones in SuperSTAR-type accelerometers, which allows a high accuracy measurement along and around all three axes.To reach the technical and performance needs of NGGM, improvements in all technological fields are fundamental. In this presentation, the current development of MicroSTAR with its challenges and key strengths will be presented.

DOI: 10.5194/gstm2026-35 · https://doi.org/10.5194/gstm2026-35

GSTM2026-36 B.2 · CRYOSPHERE · ORAL

Mass column height determined from a TVG field by inversion in the spatial domain

Josef Sebera and Aleš Bezděk

Astronomical Institute, ASCR, CZ Astronomical Institute, ASCR, CZ

Time-variable gravity (TVG) fields from satellite gravimetry missions deliver water column height by employing the spectral approach that relates the gravitational potential and the mass column height (i.e., equivalent water height, EWH). Such EWH can be rescaled at a later stage by a density ratio to adapt the column heights to a lateral density variation. Mass column height is important when satellite gravimetry products are combined/compared with geometrical techniques such as satellite altimetry. Checking this rescaling is thus desirable as the gravimetry and altimetry can only be put together by appropriate density modelling (apart from other questions such as temporal and spatial resolution etc.). Validation of the spectral approach and the rescaling can be done only by an independent approach. In our contribution, we show that it can be done by inverting TVG fields in the spatial domain if necessary corrections are also applied in the spatial domain. The spectral and spatial approaches for estimating mass column height differ in many respects, each offers certain pros and cons, e.g. the spectral approach is very time effective while the spatial one offers a certain degree of freedom in its setup. The numerical examples are shown for Greenland where the TVG signal (e.g., the trend) is known to be largest and where the satellite altimetry is used with the GRACE and GRACE-FO data products.

DOI: 10.5194/gstm2026-36 · https://doi.org/10.5194/gstm2026-36

GSTM2026-37 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Assessing the potential of future gravity missions to monitoring the Atlantic Meridional Overturning Circulation

Rory Bingham1 and Chris Hughes2

School of Geographical Sciences, University of Bristol, United Kingdom of Great Britain – England, Scotland, Wales · School of Environmental Sciences, University of Liverpool, United Kingdom of Great Britain – England, Scotland, Wales

The Atlantic Meridional Overturning Circulation (AMOC) is a major component of the climate system, redistributing heat, freshwater and carbon through the Atlantic. Although substantial weakening is projected under continued climate change, the magnitude and timing of future change remain uncertain. Detecting a sustained anthropogenic trend is particularly difficult because direct observations are spatially limited and short relative to the AMOC's strong natural variability. Satellite gravimetry offers a complementary observing strategy because AMOC variability produces a characteristic ocean-bottom-pressure (OBP) signal across the western boundary of the North Atlantic. However, the narrow cross-slope scale of this signal, together with measurement noise and leakage from continental hydrology, presents a demanding observational challenge. We assess the feasibility of recovering AMOC variability and long-term change using simulated gravimetry observations. Monthly output from a 1/12° North Atlantic ocean model is used to establish the relationship between AMOC transport and OBP over the upper and lower continental slope. Modelled OBP fields are transformed into spherical harmonics and progressively truncated to quantify the effects of limited spatial resolution. We then apply simulated mission errors and filtering representative of GRACE-C-like, Next Generation Gravity Mission (NGGM) and MAGIC configurations. Spherical-harmonic truncation substantially attenuates the narrow western-boundary OBP signals, but their cross-slope contrast, and hence the associated AMOC variability, remains recoverable at resolutions achievable by future missions. The most robust estimate is obtained from the difference between lower- and upper-slope OBP averaged over carefully selected latitude ranges. GRACE-C-like errors are generally too large for reliable recovery, whereas NGGM and MAGIC reproduce much of the modelled interannual-to-decadal variability; MAGIC provides a smaller additional improvement over NGGM alone. Sustained trends are also recoverable over plausible mission lifetimes, with trend uncertainties of approximately 0.8 Sv decade^{-1} and weakening rates of around 2 Sv decade^{-1} or greater becoming detectable in the simulations. These results demonstrate that next-generation satellite gravimetry could provide basin-scale monitoring of AMOC variability and emerging trends, complementing sustained in-situ arrays and extending observations beyond a single latitude.

DOI: 10.5194/gstm2026-37 · https://doi.org/10.5194/gstm2026-37

GSTM2026-38 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Directional gradient regularization of gravity field solutions

Roelof Rietbroek and Sedigheh Karimi

University of Twente, Faculty of Geo-information and Earth Observation, Enschede, Netherlands University of Twente, Faculty of Geo-information and Earth Observation, Enschede, Netherlands

The spurious North-South stripes in gravity field solutions of satellite gravimetry mission such as GRACE and its follow-on (GRACE-FO), are a well known issue. They are thought to be the result of residual non-Gaussian errors, most notably stemming from residual sub-monthly dealiasing model errors, which propagate into the gravity field solutions. The typical pattern is known to be linked to the orbit geometry (e.g. Peidou and Pagiatakis 2020) in combination with the directional sensitivity of the raw observations.

Over the years, various approaches have been developed to reduce these striping patterns as a trade-off against signal attenuation, resulting in varying filtered solutions. One particular type of these approaches stems from using diagonal regularization matrices using approximate or real error-covariance matrices (e.g. Kusche et al 2009, Klees et al 2008, Horvath et al. 2018). These can be done either at the normal equation system level or be formulated as a post-processing filter operator.

In this work, we revisit this regularization principle and explore ways of constraining the East-West and North-South gravity field gradients at given satellite heights.

We show that, in the spherical harmonic domain, the associated regularization matrices resolve to order-block-diagonal matrices with a typical checkerboard pattern separating even and odd degrees.

Furthermore, we compute an empirical degree varying power law for the gradients based on the ESA earth system model (Dobslaw et al 2015). We present the result of testing various regularization strengths and weighing schemes, which are applied to the publicly available normal equation system from ITSG. The regularized solutions are compared against conventional filter approaches (DDK filters, Gauss,) in the spatial and spectral domain.

DOI: 10.5194/gstm2026-38 · https://doi.org/10.5194/gstm2026-38

GSTM2026-39 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Optimized SLR mass change and degree 1 estimation with a genetic algorithm

Bryant Loomis1, Terence Sabaka1, Kenneth Rachlin2, Michael Croteau1, and Frank Lemoine1

NASA GSFC, Greenbelt, United States of America · KBRwyle, Greenbelt, United States of America

Satellite laser ranging (SLR) tracking data is routinely used to determine low-degree time-variable gravity fields, typically to maximum degrees/orders ranging from 5 to 10. Several methods have been developed to combine the longer and continuous record of SLR and the significantly better spatial resolution and accuracy of the GRACE and GRACE-FO fields to construct gap-free mass change estimates starting in 1993 (or earlier), each with their own limitations and assumptions. Our group has previously shown that large-scale regularized mascons can be estimated from SLR tracking data, while also noting that these initial solutions exhibited high sensitivity to the regularization design. Here we present a new set of large global mascons derived from regularized SLR inversions, where the regularization matrices (diagonal signal covariance matrices centered on a zero state) have been optimized using a simple genetic algorithm. The performance metric (i.e., fitness) of each randomized mutation is defined as the misfit between the de-trended SLR and GRACE/-FO solutions, noting that at these large spatial scales GRACE/-FO mass changes can safely be treated as the truth. This work results in a unique continuous time series of global mass change, including degree 1, from 1993 to the present. We compare our results to GRACE/-FO regional mass change and to IMBIE assessments of ice sheet mass change.

DOI: 10.5194/gstm2026-39 · https://doi.org/10.5194/gstm2026-39

GSTM2026-40 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

ODIN and THOR: optomechanical inertial sensors for space geodesy

Felipe Guzman and Jose Sanjuan

University of Arizona, Wyant College of Optical Sciences, Tucson, United States of America University of Arizona, Wyant College of Optical Sciences, Tucson, United States of America

ODIN and THOR represent two complementary advances in low-frequency optomechanical inertial sensing aimed at enabling next-generation mass change missions. ODIN, the Optomechanical Distributed Instrument for Inertial Sensing and Navigation, is a compact, low-cost, and power-efficient system currently supported by NASA's InVEST program and manifested for flight on the GRATTIS mission in February 2027. It consists of an array of linear accelerometers capable of high-precision inertial measurements below 10^{-9} m $s^{-2}/\sqrt{Hz}$, with angular sensitivities better than 50 µrad/√Hz. Laboratory demonstrations have reached noise floors below 8 pico-g above 60 mHz, positioning ODIN as a strong candidate for mass change studies, inertial navigation in GPS-denied environments, and planetary seismology, where size, weight, and power constraints dominate.THOR, the Triaxial High-sensitivity Optomechanical Reference, builds directly on ODIN heritage and is aimed at maturing optomechanical triaxial sensing technologies for spaceborne mass change observations. THOR targets linear acceleration sensitivities of 10^{-9} m $s^{-2}/\sqrt{Hz}$ along three orthogonal axes across a measurement bandwidth of 1–100 mHz.Together, ODIN and THOR

advance optomechanical inertial sensing toward higher technology readiness, offering reduced cost, size, weight, and power, an innovative alternative to electrostatic systems, and support for flexible mission architectures. I will present the current status of these two spaceflight instruments.

DOI: 10.5194/gstm2026-40 · https://doi.org/10.5194/gstm2026-40

GSTM2026-41 B.3 · OCEANOGRAPHY · ORAL

Deployment of a Long-Term Self-Calibrating Ocean Bottom Pressure Recorder in the Nordic Seas for GRACE-FO, GRACE-C, and NGGM Validation

Cecilia Peralta-Ferriz, John Guthrie, and James Morison

Applied Physics Laboratory, University of Washington, Seattle, USA Applied Physics Laboratory, University of Washington, Seattle, USA

We report the successful deployment of a Sonardyne Fetch AZA ocean bottom pressure recorder at the Oceanographic Weather Station M (66°N, 2°E) in the Nordic Seas in June 2026. The deployment and data recovery efforts are supported by NASA's GRACE-FO Science Team and the University of Bergen, with the goal of providing an independent, long-term in situ record for the validation and interpretation of satellite gravimetry observations. The instrument was deployed at approximately 2,034 m depth, about 2 km west of the long-term physical oceanographic mooring at Station M, and is recording hourly ocean bottom pressure and temperature. The instrument is equipped with batteries designed to support up to 10 years of operation and an acoustic transducer that enables data to be retrieved remotely from the surface without recovering or disturbing the instrument on the seafloor. A key capability of the Fetch AZA is its in situ self-calibration system, which enables characterization and correction of long-term pressure-sensor drift – a major limitation of conventional bottom pressure records for resolving low-frequency ocean mass variability. An initial data harvest approximately 11 hours after deployment successfully recovered the first measurements and confirmed proper instrument operation. Strong acoustic communications were also established from multiple ranging locations approximately 1 km from the triangulated instrument position, demonstrating the feasibility of future acoustic data recovery. Annual data recovery is planned to build a multi-year, drift-corrected ocean bottom pressure record for comparison with GRACE-FO and future GRACE-C and NGGM observations, and for validation and improvement of their data products.

DOI: 10.5194/gstm2026-41 · https://doi.org/10.5194/gstm2026-41

GSTM2026-42 B.4 · HYDROLOGY · ORAL

Enhanced Retrieval of the Seasonal Cycle of Terrestrial Water Storage

Matthew Rodell1, Mary Michael O'Neill1,2, and Bryant Loomis1

Earth Sciences Division, NASA Goddard Space Flight Center, Greenbelt, MD, USA · Earth System Science Interdisciplinary Center, University of Maryland, College Park, MD, USA

Standard spherical harmonic and mascon approaches can attenuate mass change signals. Among other effects, this leads to mischaracterization of the seasonal cycle of terrestrial water storage (TWS), spatially smoothing and underestimating the peak amplitude in particular. When GRACE/FO derived TWS is included in a water budget analysis, that underestimation leads to a mismatch with the accumulated fluxes of precipitation, evapotranspiration, and runoff. Herein we present results of a new approach to deriving the seasonal cycle of TWS, which relies on L1B data stacking. We show that the new technique better retains the amplitude of the seasonal signal, making it more appropriate for use in water budget analyses.

DOI: 10.5194/gstm2026-42 · https://doi.org/10.5194/gstm2026-42

GSTM2026-44 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · POSTER

Stochastic modeling of non-tidal atmospheric and oceanic dealiasing models for GFZ GRACE/GRACE-FO Level-2 processing

Michael Murböck1, Christoph Dahle2, Natalia Panafidina2, Markus Hauk2, Josefine Wilms2, Karl Hans Neumayer2, and Annette Eicker2

Technische Universität Berlin, Weßling, Germany · GFZ German Research Centre for Geosciences, Global Geomonitoring and Gravity Field, Weßling, Germany

The central hypothesis of the Research Unit (RU) New Refined Observations of Climate Change from Spaceborne Gravity Missions (NEROGRAV) reads: only by concurrently improving and better understanding of sensor data, background models, and processing strategies of satellite gravimetry, the resolution, accuracy, and long-term consistency of mass transport series from satellite gravimetry can be significantly increased; and only in that case, the potential of future technological sensor developments can be fully exploited.In continuation of the first RU phase, the individual project Improved Stochastic Modeling in GRACE/GRACE-FO Real Data Processing (ISTORE-2) aims to complete the optimized stochastic modeling for GRACE and GRACE-FO gravity field determination. This includes stochastic modeling of the non-tidal atmospheric and oceanic dealiasing (AOD) models which were recently implemented into the GRACE/GRACE-FO Level-2 processing at the GFZ Helmholtz Centre for Geosciences. In this context, we co-estimate AOD model coefficients using AOD error variance-covariance matrices (VCMs) in terms of constraint matrices.This presentation provides an overview of the main processing steps together with AOD error analyses. In particular, we investigate the impact of taking into account not only stationary but also non-stationary AOD VCMs. Results are presented in terms of gravity field solutions in the spectral and spatial domain.

DOI: 10.5194/gstm2026-44 · https://doi.org/10.5194/gstm2026-44

GSTM2026-45 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Ground experiment to study the interaction of a bounded plasma with a metal shield

Anja Schlicht

TU Munich, FESG, Munich, Germany TU Munich, FESG, Munich, Germany

Unknown effects heavily disturb space accelerometers of gravity field missions. Disturbances in GRACE called twangs could be related to excitations in the ionosphere. This motivated the study of possible interaction between the ionosphere and space instruments. We report on a ground experiment conducted in a plasma chamber, where we reproduced the switching of heaters and the reaction of a shielded and un-shielded receiving antenna. We identified the signature of plasma excitations as typical for a bounded plasma in a cavity, where a neutral layer establishes between the metal surface and the plasma. In addition, we see excitations of smaller frequencies, which would fit to mechanical resonances of the metal shield. The plasma excitations could be shielded, whereas the lower frequencies stay in the response of the receiver. Beneath the presentation of the experiment, we will propose a possible mechanism for the excitation of mechanical resonances of metals by plasma waves disturbing the layer between a metal and a plasma.

DOI: 10.5194/gstm2026-45 · https://doi.org/10.5194/gstm2026-45

GSTM2026-46 B.4 · HYDROLOGY · ORAL

The Response of GRACE/GRACE-FO to 14,000 Global Lakes from 1992 to 2026

Kevin Gaastra, Donald Argus, Felix Landerer, David Wiese, and Athina Peidou

Jet Propulsion Laboratory Jet Propulsion Laboratory

We estimate changes in lake levels and volumes from October 1992 to April 2026 each month for 14,071 natural lakes and artificial reservoirs larger than 10 km2 contained within the surface water and ocean topography (SWOT) mission's prior lake database (PLD). We use altimetric lake level estimates from G-REALM, DAHITI, HydroWeb, the SWOT mission, and the Global Lake Water Storage dataset's hypsometric relationships to establish a complete timeseries of lake water storage over the study interval interpolating and extrapolating a seasonal where data is missing. From this information we construct three data products to disaggregate global lake water from GRACE terrestrial water storage: 1) timeseries of lake level and storage over the study interval, 2) maps of change in lake water for the 11,386 natural lakes and 2,685 artificial reservoirs on a 16th degree global grid each month, 3) lake-generated changes in equivalent water thickness as seen by GRACE for both JPL's 3-degree mascon solution and for the spherical harmonic solutions of GRACE. Of the 1706 3-degree terrestrial mascons 1109 (65%) are affected by lake water; 808 mascons (73%) experience a decrease in weighted root mean square (WRMS) when lake water is removed, while 301 mascons (27%) experience an increase in WRMS. This dataset shows a 14-fold increase in the number of lakes studied compared to our previous lake model as well as an extension to 2026 (Gaastra et al. 2025). Correspondingly the number of mascons affected roughly doubles and the percentage of mascons experiencing an increase in WRMS has increased as coverage of high latitude lakes is improved. High latitude and high elevation regions experience an increase in WRMS when lake water is removed primarily due to glacial and snow melt fed lakes which experience their maximum seasonal water during the summers when terrestrial water storage is low as snow has melted. We find the regions where lake water disagrees substantially with GRACE/GRACE-FO terrestrial water are mainly concentrated in high latitude regions, alpine regions, and arid near costal mascons.

DOI: 10.5194/gstm2026-46 · https://doi.org/10.5194/gstm2026-46

GSTM2026-47 B.4 · HYDROLOGY · POSTER

Quantifying total drainable water storage and its distribution using GRACE/GRACE-FO

Alireza Sobouti1, Mohammad J. Tourian1, Omid Elmi1, Benjamin Kitambo1,2, Peyman Saemian1, John T. Reager3, Fabrice Papa4, and Nico Sneeuw1

University of Stuttgart, Institute of Geodesy, Stuttgart, Germany · Congo Basin Water Resources Research Center (CRREBaC) & the Regional School of Water (ERE), University of Kinshasa (UNIKIN), Kinshasa, Democratic Republic of the Congo · NASA Jet propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA · Laboratoire d'Etudes en Géophysique et Océanographie Spatiales (LEGOS), Université de Toulouse, CNES/CNRS/IRD/UT, Toulouse, France

Total Drainable Water Storage (TDWS) is the portion of basin water storage that can be released through drainage and sustain river discharge. River discharge has intrinsic environmental value, as it provides the water needed to sustain river ecosystems, particularly during dry periods. A consistent global estimate of TDWS would therefore provide a new basis for comparing the capacity of river basins to sustain discharge and for identifying regions with comparatively limited drainable water-storage capacity. Despite its hydrological relevance, TDWS has not yet been estimated at the global scale from satellite gravimetry observations. As demonstrated in previous studies, TDWS can be inferred by linking temporal variations in basin water storage to river discharge through a runoff–storage relationship. GRACE and GRACE Follow-On (GRACE/GRACE-FO) provide a unique observational basis for this purpose by measuring Terrestrial Water Storage Anomalies (TWSA) across large river basins worldwide. However, TWSA includes storage components with different timing and routing characteristics relative to river discharge, which can introduce phase shifts and hysteresis into the storage–discharge relationship. We therefore derive Drainable Water Storage Anomalies (DWSA) by accounting separately for snow, lake, wetland, and reservoir storage.

The resulting DWSA time series are combined with monthly river discharge observations from the Global Runoff Data Centre (GRDC) to characterize the runoff–storage relationship and estimate both TDWS and the characteristic drainage time, representing the timescale over which drainable storage is released through river discharge. Here, we provide the first global basin-scale quantification of TDWS from GRACE/GRACE-FO observations by applying this framework to more than 200 globally distributed river basins using three independent GRACE/GRACE-FO mascon solutions from CSR, JPL, and GSFC. Basin-scale TWSA and the corresponding DWSA are derived independently for each mascon product, enabling us to assess the consistency and robustness of TDWS and drainage-time estimates with respect to differences in gravity-field processing. We further examine the global spatial variability of both quantities and evaluate the suitability of the runoff–storage relationship for individual basins. This evaluation is important because uncertainties in river discharge, and weak or nonlinear storage–discharge behavior can affect the reliability of the inferred basin parameters. Together, TDWS and characteristic drainage time provide complementary information on how much water a basin can release through drainage and the timescale over which this storage is discharged. Their global estimation enables a consistent comparison of drainage behavior across river basins and helps identify differences in their capacity to sustain river flow during periods without precipitation.

DOI: 10.5194/gstm2026-47 · https://doi.org/10.5194/gstm2026-47

GSTM2026-48 B.5 · MULTIDISCIPLINARY SCIENCE · ORAL

GRACE/GRACE-FO radio occultations for weather forecast and climate research

Torsten Schmidt, Patrick Schreiner, and Jens Wickert

GFZ Helmholtz Centre for Geosciences, Potsdam, Germany GFZ Helmholtz Centre for Geosciences, Potsdam, Germany

The major objective of the GRACE Follow On (GFO) mission with its two satellites GF1 and GF2 is to obtain precise global and high-resolution models for both the static and the time variable components of the Earth's gravity field. Additional goal is the continuation of the GPS radio occultation (RO) measurements from the predecessor GRACE, successfully performed between 2006 and 2017.Starting with the CHAMP mission, the GFZ has been continuously processing radio occultation data since 2001 and has been providing it operationally in near real time to weather services since 2006. GFZ RO data, as GRACE and GFO also contribute to the global RO data base for climate research.Following the successful RO missions CHAMP (2001–2008) and GRACE (2006–2017), the GFZ is currently still processing RO data from TerraSAR-X (since 2008), Tandem-X (since 2011), and the two GRACE-FO satellites (since 2020), with TerraSAR-X representing the longest continuous RO data set to date. The complete RO data set has been reprocessed since 2026.In this presentation, we discuss climatologies and temperature trends in the upper troposphere and lower stratosphere (5-40 km) and tropopause parameters derived from the GFZ data set (2002-2025) and compare the results with UCAR and ERA5 data. The importance of smaller RO missions for weather and climate research is emphasized.

DOI: 10.5194/gstm2026-48 · https://doi.org/10.5194/gstm2026-48

GSTM2026-49 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Time-Variable Gravity Recovery from Clock Frequency Comparisons in Formation-Flying Missions

Mattia Imperato1,2, Khosro Ghobadi-Far2, and R. Steven Nerem2

University of Naples Federico II, Aerospace Engineering, Italy · University of Colorado Boulder, Aerospace Engineering, United States of America

Recent advances in optical clock technology offer the possibility of directly sensing differences in the Earth’s gravitational potential through frequency comparisons between satellites. This study investigates the performance of different satellite formation-flying architectures for recovering the time-variable gravity field using optical clock measurements.Four formation geometries are considered: a GRACE-like along-track configuration and three multidirectional formations, namely Cartwheel, Pendulum, and Helix. We performed a simulation analysis over 2016-2018 and synthesized the observable, geopotential difference, based on Hydrology, Icesheet and Solid Earth

components of the v3 of the ESA Earth System Model. The observations were then contaminated with AOD1B error (from AOe07) and various clock noise levels from 10-17 to 10-20. We estimated monthly spherical harmonic solutions to degree 60, and evaluated the results in the spectral and spatial domain, as well as through examining timeseries of terrestrial water storage changes and icesheet mass change. The results show that formation geometry strongly influences the quality and spatial characteristics of time-variable gravity solutions. Our results show that a clock noise level of 10-19-10-20 is required to reliably recover time-variable gravity fields. Compared with the GRACE-like formation, multidirectional formations such as the Pendulum provide substantially improved solutions, both in terms of error magnitude and isotropy, making the errors easier to mitigate with a simple Gaussian filter. The performance of formation-flying missions depends strongly on clock noise, highlighting an important interplay between measurement precision and formation geometry. As clock performance improves, the influence of constellation geometry becomes increasingly important, with the Pendulum and Helix configurations showing particularly promising time-variable gravity recovery capabilities, especially at smaller spatial scales.

DOI: 10.5194/gstm2026-49 · https://doi.org/10.5194/gstm2026-49

GSTM2026-50 B.5 · MULTIDISCIPLINARY SCIENCE · POSTER

Enhancing the Spatial Resolution of Monthly Gravity Solutions Derived from SLR and DORIS

Adrien Joliat1,2, Junyang Gou2, Ellen Poli2, Devis Tuia1, and Benedikt Soja2

Environmental Computational Science and Earth Observation Laboratory, EPFL · Institute of Geodesy and Photogrammetry, ETH Zurich

Monthly gravity-field solutions derived from satellite laser ranging (SLR) and Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS) provide an observational record that extends well before the GRACE era but resolve only a limited range of spatial wavelengths. In this study, we implement deep learning methods to generate monthly gravity fields up to degree 60 from SLR/DORIS products (up to degree 10) by targeting COST-G GRACE(-FO) solutions. The proposed framework combines coefficient-level tokenization, a Transformer encoder, a degree-wise autoregressive gated recurrent unit (GRU) decoder, temporal conditioning, and a task-specific high-degree low-order loss. The final product is obtained by ensembling 20 independently trained models; its performance is evaluated on the held-out 2022--2023 test period. When assessed against COST-G ground truth in terms of equivalent water height, the ensemble achieves an area-weighted root-mean-square error (RMSE) of 4.06 cm and a spatial grid correlation of 0.986. It reduces grid-wise RMSE by 28.4\% relative to a recent SLR/DORIS reconstruction and achieves the best results for seven of the eight principal spectral, spatial, and basin-scale metrics. Application to the complete 1984--2023 SLR/DORIS record yields a consistent GRACE-like gravity field variation product and offers the potential to extend multiple downstream applications back through time.

DOI: 10.5194/gstm2026-50 · https://doi.org/10.5194/gstm2026-50

GSTM2026-52 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Localized Estimation of Tide Model Error and Mass Anomalies from GRACE/GRACE-FO Inter-Satellite Ranging

Nicholas Childress, Srinivas Bettadpur, Himanshu Save, Geethu Jacob, and Benjamin Krichman

University of Texas At Austin, Center for Space Research, Orbital Mechanics, Austin, United States of America University of Texas At Austin, Center for Space Research, Orbital Mechanics, Austin, United States of America

Despite improvements from the GOT5.6 ocean tide model adopted for RL07 processing, tide model error remains a significant source of temporal aliasing in GRACE gravimetry. This effect is felt particularly in shallow shelves with complex bathymetry and a larger than normal presence of nonlinear tides, as well as in regions where hydrodynamic models are least constrained by observations. The cornerstone altimetry missions used to constrain modern tide models rely on an open surface and are inclined at around 66 degrees, limiting their coverage and leaving high latitudes constrained primarily by in-situ measurements, high-inclination altimetry and hydrodynamic models.

SWOT extends coverage to 78 degrees latitude, but remains dependent on an open surface and its record since 2022 falls short of separating several closely spaced tidal constituents. GRACE and GRACE-FO have effectively global coverage, an extended time series history, and observe mass distribution rather than surface level, making them uniquely suitable for tidal estimation in high-latitude and ice-covered regions despite GRACE's relatively poor spatial resolution.We utilize a localized formulation following Han et al. (2008) to estimate residual mass anomalies and tides from GRACE inter-satellite range-rate residuals. Surface mass is parameterized as equivalent water height on a regional grid, gravitational acceleration is calculated by point-mass approximation and integrated along the nominal orbits, and mass anomaly and tidal signals are recovered relative to the CSR background gravity field and ocean tide model. Per-pass state deviation vectors are estimated to absorb out-of-region errors, long wavelength orbit signals, and accelerometer errors. The temporal basis is evaluated at each observation epoch, allowing for simultaneous estimation of sub-daily tidal harmonic signals alongside mean mass anomalies over arbitrary averaging intervals and reducing the leakage of non-tidal signals into the tidal residual estimates.The regional mass anomaly inversion of GRACE residuals is ill-posed and requires an extensive regularization scheme to produce meaningful results. Range-rate data constrain an along-track projection of the mass field, and arc state deviation parameters often compete with the mass anomaly parameters for the same long-wavelength signals. We discuss the conditioning of the problem, the regularization strategies that yielded a stable solution in the current model, and the sensitivity of the recovered field to regularization weights.Validation of the current model against CSR global solutions for monthly mean mass anomalies confirms the estimator geometry, regularization, and arc parameterization. We then present preliminary residual tide estimates in areas of interest, such as the Weddell Sea, assessed against available bottom pressure records, with proposed extensions to other regions and potential improvements to the localized mass anomaly estimation scheme.

DOI: 10.5194/gstm2026-52 · https://doi.org/10.5194/gstm2026-52

GSTM2026-53 B.2 · CRYOSPHERE · ORAL

Spatiotemporal Validation of Mass-Change Products over Greenland and Antarctica Using GRACE-FO Laser Ranging Instrument

Evan Wilson1,2,3, Khosro Ghobadi-Far1,2, and R. Steven Nerem1,2,3

Department of Aerospace Engineering Sciences, University of Colorado Boulder, CO USA · Cooperative Institute for Research In Environmental Sciences, University of Colorado Boulder, CO USA · Colorado Center for Astrodynamics Research, University of Colorado Boulder, CO USA

Observations of mass changes in the cryosphere are critical to understanding the impact of climate change and constraining global projections of sea-level rise. Observations from satellite gravimetry missions, satellite altimetry, and Surface Mass Balance (SMB) models provide differing estimates of mass change at different temporal and spatial scales making intercomparison challenging. GRACE-FO is equipped with two sensors to track the inter-satellite distance; a baseline K-band microwave ranging system (KBR) and an experimental Laser Ranging Interferometer (LRI). The improved ranging precision of the LRI enables Line-of-Sight Gravity Difference (LGD) observations with substantially greater sensitivity to short-wavelength gravity signals than is possible with the KBR and at a range of temporal scales. Building on previous work demonstrating enhanced high-frequency sensitivity over Greenland and Antarctica, we apply LGD as a validation tool for GRACE-FO mascon products, ICESat-2-derived mass-change estimates, and two versions of the RACMO SMB model over the same region. This tests products commonly used for mass-change observations in the cryosphere over a range of spatial and temporal scales. LRI-derived LGD uniquely enables an independent direct frequency domain validation of the effective gravity signals from different mas change products without the limiting effects of temporal aliasing. By forward modeling gravity from monthly mass-change solutions and comparing the resulting synthetic LGD with LRI LGD observations, we assess the ability of different mass-change products to recover signals over multiple spatial and temporal scales. The results demonstrate that the effective spatial resolution of mass-change products is not characterized by a single spatial scale but instead depends on the interaction between the magnitude of the mass-change signal, the spatial and temporal scale of mass changes, and the measurement or modeling technique. This validation is particularly important for the cryosphere where understanding the effective spatial and temporal resolution of products can change the interpretation of the data.

Validation of mass-change products using LGD provides a new approach for evaluating existing mass-change products in Greenland and Antarctica as well as guiding the development of future mass-change products.

DOI: 10.5194/gstm2026-53 · https://doi.org/10.5194/gstm2026-53

GSTM2026-54 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Assessing Machine Learning for Automated Screening of GRACE-C Quicklook Gravity Solutions

Anakin Martinez

University of Texas, Center for Space Research, United States of America University of Texas, Center for Space Research, United States of America

Scheduled for launch in 2028, GRACE-C is expected to provide Quicklook gravity field solutions with a latency of approximately three days. Such low latency necessitates an operational validation schedule best suited to automated or semi-automated frameworks. Machine learning (ML) offers the potential for rapid, automated quality screening of solutions as "physically consistent," "sub-optimal," or "anomalous." However, predicting Liquid Water Equivalent (LWE) values at the per-pixel level is challenging given the limited 241-month time series and the complex spatial, temporal, annual, and spherical correlations within the data. Additionally, month-to-month LWE variability increases throughout time, indicating increasingly dynamic hydrological signals across the GRACE and GRACE-FO eras.

To investigate the feasibility of ML to support automated validation of GRACE-C Quicklook solutions, we systematically compared a diverse set of ML architectures including regularized linear models, tree-based ensembles, spatiotemporal dimensionality-reduction techniques, and deep neural networks. Model skill is then assessed relative to a per-pixel persistence baseline, providing a consistent reference for determining actual predictive skill. We further examine the sensitivity of model performance to hyperparameters and feature selection through systematic spatiotemporal ablation experiments, with particular attention to the characteristics of GRACE-derived signals. Hydro-meteorological variables are evaluated across a range of climatic domains to characterize their predictive value. We have currently enacted a 5-fold temporal cross-validation with overlap purging alongside basin-level spatial folds to prevent data leakage and spatial-temporal memorization. These experiments provide an initial assessment of the range of useful architecture and features that may serve a physically informed approach to screening GRACE-C Quicklook solutions.

DOI: 10.5194/gstm2026-54 · https://doi.org/10.5194/gstm2026-54

GSTM2026-55 B.3 · OCEANOGRAPHY · ORAL

Seismic Effects of Large Earthquakes on Sea-Level Budget Closures

Tony Song1, Felix Landerer1, Benjamin Hamlington1, Hyeonsoo Cha2, and Jae-Hong Moon2

Jet Propulsion Laboratory, Pasadena, United States of America · Jeju National University, South Korea

Large undersea earthquakes can distort ocean bottom pressure (OBP) measurements from the GRACE and GRACE-FO satellites, making it difficult to close regional sea-level budgets in earthquake-prone areas. In this talk, we will show that these earthquake-related signals can be effectively removed using an iterative Empirical Orthogonal Function (EOF) method that captures both sudden co-seismic changes and longer-lasting post-seismic effects. The results demonstrate that these seismic signals are non-negligible and introduce a systematic bias associated with ongoing tectonic motion. In some regions, this bias can exceed sea-level changes caused by ocean warming (steric effects), while globally it can even exceed the signal expected from Glacial Isostatic Adjustment (GIA). These findings highlight the importance of accounting for seismic effects when assessing regional and global sea-level budgets and separating natural and anthropogenic contributions to observed climate change.

DOI: 10.5194/gstm2026-55 · https://doi.org/10.5194/gstm2026-55

GSTM2026-56 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Sub-monthly Terrestrial Water Storage Change from GRACE-C and NGGM: Assessing MAGIC's Capability for Rapid Mass Change Monitoring

Khosro Ghobadi-Far

University of Colorado Boulder, Department of Aerospace Engineering Sciences, United States of America University of Colorado Boulder, Department of Aerospace Engineering Sciences, United States of America

By mitigating temporal aliasing errors caused by unresolved high-frequency atmospheric and oceanic mass variations, MAGIC is expected to provide substantially improved estimates of terrestrial water storage (TWS) changes. To date, MAGIC's performance has been evaluated primarily using global gravity field inversions. In this presentation, I evaluate its performance using the Line-of-sight Gravity Difference (LGD) approach, with a particular focus on addressing two major limitations of the current GRACE/-FO missions for terrestrial hydrology: their coarse spatial resolution of ~300 km and monthly temporal sampling, which limit their ability to monitor rapidly evolving events such as floods. To this end, I perform a regional inversion of sub-monthly (10-day) surface water storage changes over the Amazon Basin. The Amazon's large-amplitude, localized hydrologic signals and rapid sub-monthly variability provide an ideal test case for evaluating spatial resolutions finer than the ~300 km achievable with monthly GRACE/-FO solutions. The results show that, for 10-day TWS change solutions, GRACE-C and NGGM achieve comparable performance at a spatial scale of 330 km, whereas NGGM outperforms GRACE-C at 220 km and 165 km. NGGM recovers approximately 80% and 65% of the 10-day TWS signal at spatial scales of 220 km and 165 km, respectively, representing a 15-20% improvement over GRACE-C. I also quantify the effects of orbital geometry (i.e., the inclined orbit) and altitude on NGGM's improved results. Between latitudes ±70°, the improved performance of MAGIC is largely driven by the contribution of the inclined NGGM pair. These results demonstrate the potential of MAGIC to substantially advance spatiotemporal monitoring of TWS change and enable observations of hydrologic processes at spatial and temporal scales beyond the capabilities of the current GRACE/-FO missions.

DOI: 10.5194/gstm2026-56 · https://doi.org/10.5194/gstm2026-56

GSTM2026-57 B.3 · OCEANOGRAPHY · ORAL

Is the contribution of glaciers and ice sheets to sea level rise slowing down?

Isabella Velicogna1,2, Arnav Agrawal1, Tyler Sutterley3, and Geruo Aa1

University of California Irvine, Earth System Science, Irvine, CA, United States of America · Jet Propulsion Laboratory, Caltech, Pasadena, CA, United States of America · University of Washington, Applied Physics Lab, Seattle, WA, United States of America

We report on the ice mass loss of the Antarctic and Greenland Icesheets and of the world Glaciers and ice caps from 2002 to 2026 using data from the Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GRACE-FO) missions.In Greenland, we detect a slowdown in ice mass loss since 2012 due to lower melt runoff production caused by a succession of colder summers, especially in the southwest and northeast sectors. The mass loss averages 245+/-47 Gt/year for the period 2002-2026 with the acceleration in loss negligible after 2012 due to reduced surface melt, especially in the southwest and northeast. A disproportionate portion of the mass loss is driven by North Greenland which includes the two largest marine-based basins, which are most prone to rapid sea level rise.In Antarctica, we detect a slowdown in ice mass loss in 2019-2024 which was caused by an increase in accumulation of snowfall, especially in Queen Maud Land, which is unprecedented in the GRACE record. Increase in snowfall is also observed in Totem and Moscow and Wilkins land East Antarctica and in the Antarctic Peninsula. This short-lived mass gain in 2019-2024 mirrors a similar episode in 2004–2007 both in magnitude and duration, reflecting interannual variability in surface mass balance, not a reversal of long-term trends. A disproportionate portion of the mass loss is controlled by the Amundsen Sea Embayment of West Antarctica and to a lesser extent by the Wilkes Land sector of East Antarctica. Both ice sheets continue to lose ice, with no statistically significant acceleration during the GRACE era, confirming that Antarctica's apparent recovery is a brief snowfall-driven pause within a persistent long-term decline.For the world Glacial and Ice Caps (GIC) we find a sustained mass loss of 296+/-30 Gt/year and an acceleration in mass loss of 6+/-2 Gt/year^2, the Arctic is still dominating the GIC mass loss signal and account for 71% of the total GIC mass loss. Ice sheet and Glaciers continue to have a major impact on sea level rise, the ice sheets show a relative loss in acceleration after 2012, but the longer record 1979-2025 shows an acceleration in mass loss.

DOI: 10.5194/gstm2026-57 · https://doi.org/10.5194/gstm2026-57

GSTM2026-58 B.4 · HYDROLOGY · ORAL

GRACE Reveals Contrasting Groundwater Responses to Surface-Water Shortages Across the Colorado River Basin

Esther Oyedele1, Susanna Werth1, and Grace Carlson2

Department of Geosciences, Virginia Tech, Blacksburg, United States of America · Department of Geography, Indiana University, Bloomington, United States of America

Groundwater is a critical component of water supply across the Colorado River Basin (CRB), particularly during prolonged surface-water shortages. We use GRACE/FO satellite gravimetry together with complementary hydrologic datasets to assess groundwater-storage variability across the Upper and Lower CRB from 2004 to 2023 and to quantify relationships between groundwater drought and surface-water deficits.The satellite record shows persistent groundwater depletion across the basin, with losses accelerating after 2019 and during the 2020–2022 drought. During this period, groundwater losses reached 17.5 ± 3.5 km³ in the Upper CRB, exceeding the 9.4 ± 6.5 km³ lost in the Lower CRB despite the Upper Basin's typically greater hydrologic resilience.Copula-based modeling combined with statistical inference further reveals contrasting groundwater–surface water relationships between the two subbasins. In the Upper CRB, groundwater drought probability increases strongly with surface-water deficits and exceeds 60% under moderate shortages. In contrast, parts of the Lower CRB show weaker statistical dependence, consistent with groundwater extraction operating partly independently of short-term surface-water availability under chronically limited supply.These results highlight distinct regional controls on groundwater vulnerability across the

CRB and demonstrate the value of GRACE-based monitoring combined with probabilistic statistical approaches for quantifying the likelihood of groundwater drought under surface-water stress.

DOI: 10.5194/gstm2026-58 · https://doi.org/10.5194/gstm2026-58

GSTM2026-59 B.3 · OCEANOGRAPHY · ORAL

Observing Long-Term Mass Changes in the Global Ocean and Ice Sheets from Satellite Gravimetry: 1993–2023

Yufeng Nie and Jianli Chen

Department of Land Surveying and Geospatial Science, The Hong Kong Polytechnic University, Hong Kong, China Department of Land Surveying and Geospatial Science, The Hong Kong Polytechnic University, Hong Kong, China

Global ocean mass change is a major driver of present-day global mean sea level (GMSL) rise and a key indicator of climate change. Over the past three decades, mass loss from the Greenland and Antarctic ice sheets has been a major contributor to global ocean mass change. Since 2002, GRACE/-FO has provided invaluable observations of ice-sheet and ocean mass changes, whereas direct observations remain limited for the pre-GRACE era. Here, we use time-variable gravity fields derived from GRACE/-FO, satellite laser ranging (SLR) and Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS) to estimate long-term mass changes in the global ocean and ice sheets since 1993, the beginning of the high-precision satellite altimetry era. Major challenges associated with signal leakage and geocenter motion are addressed using a global forward-modeling technique. Using these satellite-gravimetry-based estimates, we reconcile the GMSL budget and compare ice sheet mass changes with estimates from independent techniques. Our results demonstrate the added value of SLR and DORIS observations in complementing the GRACE/-FO record and highlight the importance of further investigations into historical satellite gravimetry data to improve estimates of long-term mass changes in the Earth system.

DOI: 10.5194/gstm2026-59 · https://doi.org/10.5194/gstm2026-59

GSTM2026-60 B.5 · MULTIDISCIPLINARY SCIENCE · ORAL

A Framework for the Assimilation of GRACE/GRACE-FO Observations into JPL Terrestrial Reference Frame Products

Claudio Abbondanza, Toshio M. Chin, Richard S. Gross, and Michael B. Heflin

California Institute of Technology, Jet Propulsion Laboratory, Pasadena, United States of America California Institute of Technology, Jet Propulsion Laboratory, Pasadena, United States of America

Space gravimetry from GRACE and GRACE Follow-On (GRACE/GRACE-FO; GGFO)

provides quasi-monthly estimates of global mass redistribution that can

be mapped into elastic surface displacements. Although spatially coarser

than the station-based observations underpinning terrestrial reference frame

solutions, GGFO-derived displacements show substantial consistency with GNSS

estimates, particularly in the vertical component, where correlations with

IGS Repro3 solutions approach 0.7 for both CSR and JPL mascon products. This

complementarity offers an opportunity to improve the representation of

non-tidal surface deformation in terrestrial reference frame (TRF) solutions.JPL TRF products are generated using a square-root information Kalman filter

that assimilates product-level inputs from GNSS, VLBI, SLR, and DORIS within

a common state-space framework. In JTRF2020 and subsequent realizations,

non-tidal loading is represented primarily through co-estimated seasonal and other temporal components. Here we investigate an alternative formulation in which GGFO-inferred deformation is introduced as an exogenous control input driving the load-responsive components of the TRF state, so that gravimetry contributes to state evolution rather than entering as an additional observation equation.Starting from Level-3 mascon solutions, surface mass fields are transformed into the spherical-harmonic domain and adjusted to obtain a consistent representation of time-variable gravity, including treatment of degree-1 and other low-degree coefficients, model-dependent corrections, and removed and restored non-tidal geophysical variability. The framework can also operate directly on conventional spherical-harmonic gravity solutions. Gaps in monthly geopotential-coefficient time series are reconstructed using multichannel singular spectrum analysis (MSSA), followed by interpolation to daily cadence using piecewise cubic Hermite interpolation. The resulting daily fields are converted to elastic surface displacements using load Love numbers and mapped to sites represented in the JPL frame solution.This approach also separates spatial scales within the TRF estimation problem. GGFO constrains higher-degree load-induced deformation, while satellite laser ranging remains the principal technique for determining degree-1 geocenter motion. Incorporating GGFO-derived deformation into the station-position state may therefore reduce aliasing between regional or continental loading and SLR-derived common-mode geocenter motion.We present the methodology, validation, and current status of these assimilation efforts toward a unified state-space representation of terrestrial reference frame dynamics that combines space gravimetry and geometric geodesy.

DOI: 10.5194/gstm2026-60 · https://doi.org/10.5194/gstm2026-60

GSTM2026-61 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · POSTER

Hydrological de-aliasing model data for GRACE/-FO gravity field processing

Laura Jensen1, Felix Laurenz Öhlinger2, Torsten Mayer-Gürr2, Linus Shihora1, and Henryk Dobslaw1

GFZ Helmholtz Centre for Geosciences, Department 1: Geodesy, Potsdam, Germany · Institute of Geodesy, Graz University of Technology, Graz, Austria

Temporal aliasing artefacts arising from unaccounted high-frequency mass variations in the geophysical fluid layers of the Earth are one of the most important error sources in present-day GRACE/-FO monthly-mean gravity field time-series. While non-tidal atmosphere-ocean signals are routinely corrected with the AOD1B product, there is no commonly agreed strategy to cope with aliasing effects from rapid changes in terrestrial water storage (TWS).In this contribution, we investigate the impact of introducing hydrological background model data into GRACE/-FO processing on the quality of the resulting monthly solutions. For the hydrological mass signal, we use sub-monthly TWS output from the recently improved open-source global hydrological model OS LISFLOOD, which we introduce into the ITSG-Grace processing scheme operated at TU Graz. We evaluate the effects of (i) deterministic information about TWS variability only, and (ii) additionally stochastic information on the remaining uncertainties in the

modelled TWS data, which are obtained by performing a small ensemble of OS LISFLOOD model runs forced with different atmospheric reanalysis data.Even though temporal aliasing artefacts associated with TWS variability are substantially smaller than those originating from atmosphere and ocean, we conclude that the incorporation of hydrological background model information as derived from OS LISFLOOD into GRACE/-FO processing is advantageous for the quality of monthly solutions and should be considered to be included into the next major reprocessing of the satellite data record. With the provision of the new hydrological de-aliasing product at GFZ we establish the basis for this.

DOI: 10.5194/gstm2026-61 · https://doi.org/10.5194/gstm2026-61

GSTM2026-62 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · POSTER

The updated ESA Earth System Model for Future Gravity Mission Simulation Studies: ESA ESM 3.0

Linus Shihora1, Marius Schlaak2, Volker Klemann1, Laura Jensen1, Robert Dill1, Yoshiyuki Tanaka3, Ingo Sasgen4, Bert Wouters5, Shin-Chan Han6, Jeanne Sauber7, Carla Braitenberg8, Muhammad Tahir Javed8, Gerardo Maurizio8, Hugo Lecomte9, and Henryk Dobslaw1

GFZ Helmholtz Centre for Geosciences, 1.3 Earth System Modelling, Potsdam, Germany · LEGOS, Université de Toulouse, CNES, CNRS, IRD, 31400 Toulouse, France · The University of Tokyo, Department of Earth and Planetary Science, Tokyo, Japan · Alfred-Wegener-Institut, Helmholtz-Zentrum für Polar- und Meeresforschung, Bremerhaven, Germany · TU Delft, Civil Engineering, Geoscience and Remote Sensing, Netherlands · Ohio State University, School of Earth Sciences, Columbus, OH, United States · NASA Goddard Space Flight Center, Geodesy and Geophysics Laboratory, Greenbelt, MD, United States · Università degli Studi di Trieste, Dipartimento di Matematica e Geoscienze, Trieste, Italia · Finnish Geospatial Research Institute, National Land Survey of Finland

he ESA Earth System Model (ESA ESM) provides a synthetic data set of the time-variable global gravity field that includes realistic mass variations in atmosphere, oceans, terrestrial water storage, continental ice sheets, and the solid Earth on a wide range of spatial and temporal frequencies. For more than 10 years already, it is widely applied as a source model in end-to-end simulation studies for future gravity missions, but has been also utilized to study novel gravity observing concepts on the ground. For those purposes, the ESM needs to include a wide range of signals even at very small spatial scales which might not yet have been reliably observed by any active satellite mission.In this contribution, we present the details of the newly released version 3.0 of the ESA ESM as well as the first simulation studies based on the new model. The changes to the previous ESA ESM version 2 include the utilization of a small ensemble of co- and post-seismic earthquake signals, an updated GIA model, additional mass balance signals from previously not considered Arctic glaciers, sub-monthly surface-mass balance changes and a more realistic representation of ice sheet dynamics. Extreme hydrometeorological events as well as climate-driven and anthropogenic impacts on continental water storage are represented through an update of the hydrological component. Additionally, the ESM separately includes ocean bottom pressure variations along the western slope of the Atlantic, representing variations in the meridional overturning circulation as a critically important component of the interactively coupled global climate system. ESA ESM 3.0 is available with 6-hour resolution from January 2007 until December 2020. It is augmented with synthetic error time series for atmosphere and ocean as well as hydrology to facilitate stochastical modelling of residual background model errors.

DOI: 10.5194/gstm2026-62 · https://doi.org/10.5194/gstm2026-62

GSTM2026-63 B.4 · HYDROLOGY · ORAL

Beyond GRACE: Benchmarking Global Hydrological Models Back to the 1990s with Satellite Laser Ranging

Filip Gałdyn1, Laura Jensen2, Radosław Zajdel1, Krzysztof Sośnica1, and Henryk Dobslaw2

Wrocław University of Environmental and Life Sciences, Institute of Geodesy and Geoinformatics, Wrocław, Poland · GFZ Helmholtz Centre for Geosciences

GRACE and GRACE Follow-On (GRACE-FO) provide the primary satellite gravimetry observations of terrestrial water storage (TWS) variations from 2002 onwards. However, continuously improving global hydrological models often span substantially longer periods, while their validation in the pre-GRACE era remains challenging and requires complementary observations or long-term gravity field reconstructions. The validation can be conducted by combining information from GRACE with other geodetic techniques, such as Satellite Laser Ranging (SLR) and Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS). An important question, however, is to what extent hydrological model assessments depend on the choice of long-term gravity field reconstruction used as a benchmark.To address this question, we use our recently developed long-term SLR+GRACE gravity field reconstruction, which extends GRACE information back to 1995 by combining the complementary strengths of SLR and GRACE. The lowest-degree spherical harmonics are constrained directly by SLR, while intermediate and higher degrees incorporate GRACE-derived temporal and spatial information to provide gravity field solutions up to degree and order 60 also for the pre-GRACE period.

We compare this reconstruction with an independently developed SLR+DORIS gravity field solution over the common period 1995–2022. Their applicability as long-term hydrological benchmarks is evaluated for the 100 largest river basins using TWS estimates from OS LISFLOOD and the Land Surface Discharge Model (LSDM), which has been widely used in geodetic applications for nearly two decades. Model benchmark agreement is quantified using the centered Kling–Gupta Efficiency (cKGE) and Nash–Sutcliffe Efficiency (NSE). In addition, differences in NSE between OS LISFLOOD and LSDM (ΔNSE) are compared between the SLR+GRACE and SLR+DORIS benchmarks to assess whether both gravity field reconstructions lead to the same qualitative ranking of the hydrological models.The results show that OS LISFLOOD is generally more consistent with the gravimetric benchmarks than LSDM, particularly in large tropical and subtropical basins. For the SLR+GRACE reconstruction, median cKGE values are 0.59 for OS LISFLOOD and 0.50 for LSDM for detrended TWS, and 0.41 and 0.23, respectively, for interannual variability. More importantly, SLR+GRACE and SLR+DORIS yield the same sign of ΔNSE in 70.7% of cases for detrended TWS and 64.3% for interannual variability. The median absolute difference in ΔNSE between the two benchmarks increases from 0.12 to 0.20, indicating that model assessment is largely robust to benchmark choice, but becomes more sensitive at interannual timescales.In addition, degree-1 coefficients and the corresponding geocenter motion were reconstructed from the SLR+GRACE solution for 1995–2023. This extends the reconstructed geocenter series into the pre-GRACE era, fills the gap between GRACE and GRACE-FO, and complements our long-term gravity field reconstruction with degree-1 information.

DOI: 10.5194/gstm2026-63 · https://doi.org/10.5194/gstm2026-63

GSTM2026-64 B.4 · HYDROLOGY · ORAL

Assimilation of GRACE/-FO data into global hydrological models for water use scenarios

Maike Schumacher1, Çağatay Çakan1, Supriya Tiwari1, Fan Yang1, and Ehsan Forootan1,2

Geodesy Group, Department of Sustainability and Planning, Aalborg University, Aalborg, Denmark · School of Geographical Sciences, University of Bristol, Bristol, United Kingdom

Global Hydrological Models (GHMs) are an invaluable tool for simulating the dynamics of our freshwater cycle and estimating its contribution to sea level rise. However, uncertainties of input data (e.g., meteorological forcing, water demand estimates) and empirical parameters, as well as errors in the model structure (e.g., simplified representation

through conceptual equations) lead to hydrological predictions that can vary strongly between different GHMs. Validation with independent satellite and in-situ observations shows that no single model can outperform the others in all regions and in all periods. Thus, assimilation of hydrological state variables into GHMs is a great opportunity to reduce model prediction uncertainty and improve performance statistics. Over the last two decades, the assimilation of terrestrial water storage (TWS) anomalies observed by the satellite gravity missions GRACE and GRACE-FO into land surface and water balance models has shown great potential to improve predictive capacity.In this presentation, we focus on exploiting the potential of TWS data assimilation into GHMs to reveal the human water use impact on shifts of TWS patterns under changing climate. We compare two decades of TWS simulated by the open-source and Python re-programmed WaterGAP Global Hydrology Model (WGHM) version while (a) disregarding and (b) considering surface water and groundwater extraction to isolate the human impact on the terrestrial water cycle. The identified patterns are compared to GRACE/-FO-derived TWS long-term trends to identify their correlation for regions with intense water use activities. This enables to map regions under acute and chronic water stress, where water overconsumption shows significant impact on shifting TWS patterns. In addition, we perform a global TWS data assimilation experiment into the World-Wide Water Resources Assessment (W3RA) model using the open-source Python-based Global Land Data Assimilation system (PyGLDA) developed by our research group. A comparison of the W3RA predictions before and after data assimilation reveals the spatial and temporal signatures that are introduced from GRACE/-FO observations. Comparisons to the isolated human impact on the terrestrial water cycle from WGHM simulations enables the generation of a global map displaying the potential of GRACE/-FO data for improved representation of human activities on the water cycle.

DOI: 10.5194/gstm2026-64 · https://doi.org/10.5194/gstm2026-64

GSTM2026-65 B.4 · HYDROLOGY · ORAL

Using GRACE as an Observational Constraint for Exponential Depth Extrapolation of Satellite Soil Moisture

Daniel Blank1, Daniel Rasche2, Andreas Güntner2,3, and Annette Eicker2,1

HafenCity University Hamburg, Hamburg, Germany · GFZ Helmholtz Centre for Geosciences, Telegrafenberg, Potsdam, Germany · University of Potsdam, Institute of Environmental Science and Geography, Potsdam, Germany

Changes in terrestrial water storage (TWS) can be observed globally by the Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GRACE-FO) mission, whereas microwave remote sensing primarily provides information on near-surface soil moisture. However, obtaining reliable information on water storage changes in deeper soil layers remains a challenge. A simple yet effective approach is the use of an exponential filter, which approximates deeper soil moisture by accounting for the delayed response to surface soil water variations. Yet, constraining these deeper storage dynamics at the global scale remains difficult. GRACE-derived TWS offers a global observational constraint on integrated subsurface water storage variations.This study utilises soil moisture products from the European Space Agency Climate Change Initiative (ESA CCI), including both surface and root zone soil moisture, together with daily GRACE-derived TWS to assess the potential of GRACE for constraining exponential depth extrapolation of satellite soil moisture. Focusing on detrended, deseasonalised sub-seasonal signals, we compare the agreement between daily GRACE-derived TWS and exponentially filtered soil moisture for different soil moisture integration depths. This allows us to assess how well the observed and depth-extrapolated soil moisture dynamics represent sub-seasonal TWS variability and to identify regions where storage dynamics beyond the represented soil depth may be relevant. The results demonstrate the potential of GRACE as an observational constraint for depth-extrapolating satellite soil moisture and reveal spatial differences in the relevance of deeper subsurface storage dynamics.

DOI: 10.5194/gstm2026-65 · https://doi.org/10.5194/gstm2026-65

GSTM2026-66 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Signal separation in GRACE and GRACE-FO data using a deep learning algorithm

Betty Heller-Kaikov1, Marius Schlaak2, Roland Pail3, and Benedikt Soja1

Chair of Space Geodesy, ETH Zurich, Zurich, Switzerland · Univ Toulouse, CNES, CNRS, IRD, LEGOS, Toulouse, France · Chair for Astronomical and Physical Geodesy, TU Munich, Munich, Germany

Time-variable gravity data as derived from the GRACE and GRACE-FO satellite missions contain the superposition of various mass change signals such as terrestrial water storage or ice mass change, as well as strong correlated noise. A common strategy for extracting specific geophysical signals from the data is to apply de-noising filters such as the DDK or VDK filters for suppressing the noise, and subsequently reducing non-targeted signals using physical forward models. Since both of these steps introduce and propagate errors to the extracted geophysical target signal, we propose an alternative strategy using a deep learning algorithm.The method consists of two subsequent steps: In the first step, the neural network is trained to co-predict several gravity signal components and a noise component when given their sum. For this training step, realistic forward-modeled signal components given by the ESA Earth System Model and a numerically simulated noise component assuming a GRACE-type gravity mission are used. In the subsequent testing step, the trained algorithm is applied to real GRACE/-FO monthly anomalous gravity grids. The proposed pipeline covers both the de-noising and signal separation step, and thereby represents an alternative processing method for (unfiltered) Level-2B to Level-3 GRACE/-FO data processing. Comparing the achieved results to equivalent state-of-the-art GRACE/-FO Level-3 data products such as those provided by the GFZ GravIS portal shows the proposed method to be beneficial in terms of the de-noising performance, as well as the preservation of small-scale details in the estimated signals.

DOI: 10.5194/gstm2026-66 · https://doi.org/10.5194/gstm2026-66

GSTM2026-67 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Stochastic modelling in attitude determination of GRACE-FO satellites for ITSG gravity field solutions

Gabriel Valles-Valverde, Felix Öhlinger, and Torsten Mayer-Gürr

Graz University of Technology, Geodesy, Theoretical Geodesy and Satellite Geodesy, Graz, Austria Graz University of Technology, Geodesy, Theoretical Geodesy and Satellite Geodesy, Graz, Austria

The GRACE/GRACE-FO mission has been monitoring monthly variations of the Earth’s gravity field for more than two decades. One source of error in the gravity field estimation process is the uncertainty of the satellites’ attitude. The errors of the attitude determination propagate to the inter-satellite range observations via the antenna offset centre (AOC) correction.Compared to GRACE, GRACE-FO carries an additional third star camera, which, together with the gyroscopes, usually makes attitude errors negligible. However, the current coarse pointing mode, employed to increase mission lifetime, enlarges the propagated attitude errors.As a next step, we aim to incorporate full covariances for the AOC correction into the stochastic model. For this purpose, this work conducts in-house level-1A attitude data processing, including sensor fusion of the star camera and gyroscope data, along with variance component estimation. The stochastic model of the star camera data is extended to account for spatial dependencies across the celestial sky, as well as for temporal correlations. The impact on the post-fit residuals and the quality of the recovered gravity field is analysed as part of the ITSG gravity field processing at the Institute of Geodesy at Graz University of Technology.

DOI: 10.5194/gstm2026-67 · https://doi.org/10.5194/gstm2026-67

GSTM2026-68 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

A traceable GRACE/GRACE-FO Level-1C scientific dataset (2002-2026) for

reproducible mass-change analysis

Fan Yang1, Yi Wu2, Maike Schumacher1, and Ehsan Forootan1

Aalborg University, Geodesy Group, Department of Sustainability and Planning, Aalborg, Denmark · Huazhong University of Science and Technology, Wuhan, China

While monthly Level-2 products of GRACE/-FO are widely used as essential climate variables for mass-transport studies, published processing chains and intermediate outputs are often insufficiently documented for transparent diagnostics, method intercomparison, and fully reproducible re-solving. Building on existing uses of the Level-1C concept, we present Hawk Level-1C, an integrated scientific dataset that links calibrated Level-1B observations with monthly Level-2 gravity fields and supports consistent, transparent, traceable, and reproducible mass-change studies with GRACE/GRACE-FO. Specifically, this Level-1C dataset comprises four subsets: (1) K-band range-rate residuals, (2) line-of-sight gravity differences, (3) normal equations, and (4) calibrated accelerometer data. Each subset provides much complementary information beyond conventional Level-2 gravity fields and opens up new scientific applications, making it possible to (1) recover the submonthly gravity signal, (2) monitor rapidly evolving events such as floods and hurricanes, (3) enable accurate land water assimilation/fusion, and (4) perform neutral density modeling related to space weather. This product is provided on a monthly basis, spanning the available mission months from April 2002 to June 2026, to cover both GRACE and GRACE-FO, which will be openly available through Zenodo soon. In addition, we declare that this Level-1C, as well as the companion Level-2 product, is completely reproducible through our open source software and the configuration files provided. In this work, the entire data processing chain from Level-1B to Level-1C and Level-2 is documented. Long term diagnosis and rigorous validation of our Level-1C against other publicly available resources are carried out, which have demonstrated the satisfactory quality of our product. In the end, we showcase how this product helps to explore and exploit the scientific potential of GRACE/GRACE-FO in a manner different from the standard Level-2 product.

DOI: 10.5194/gstm2026-68 · https://doi.org/10.5194/gstm2026-68

GSTM2026-69 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Benefits of future satellite gravimetry missions for characterizing extreme wet events in terrestrial water storage

Klara Middendorf1, Laura Jensen2, Marius Schlaak3,4, Julian Haas2, Henryk Dobslaw2, Roland Pail3, Andreas Güntner2, and Annette Eicker1,2

HafenCity Universität Hamburg, Geodesy and Geoinformatics, Hamburg, Germany · GFZ Helmholtz Centre for Geosciences, Telegrafenberg, Potsdam, Germany · Technical University of Munich, Munich, Germany · Université de Toulouse, LEGOS (CNES, CNRS, IRD, UT), Energie, mass et changement climatique, 18 Avenue Edouard Belin, 31401 Toulouse Cedex 9, France

Under the assumption that a warming climate leads to an intensification of the global water cycle, it is hypothesized that also the occurrence frequency and severity of extreme events such as droughts and floods will increase in the upcoming decades. GRACE/-FO observations of terrestrial water storage (TWS) have been used in the past to identify and analyse extreme events both on a global and regional scale. However, these analyses are restricted by the limited spatial and temporal resolution of current satellite gravimetry observations. Especially, flooding events tend to occur very locally and with short temporal (sub-monthly) extent, thus capturing them is challenging. Future satellite gravimetry missions, particularly the double-pair constellation MAGIC, are expected to significantly enhance the spatial and temporal resolution. In this study, we globally investigate the benefit MAGIC can achieve to detect wet extreme events using long-term (50 years) end-to-end simulations of GRACE-C and MAGIC.The simulation environment is based on the acceleration approach and considers tidal and non-tidal background model errors as well as instrument noise of the acceleration and ranging instruments following the current MAGIC mission design studies. As input and reference, we use the daily output of a climate model (GFDL-CM4) from the CMIP6 archive that has been identified as a realistic representation of water storage evolution in previous studies. To explore the improved temporal and spatial resolution expected from the MAGIC constellation, we (i) compare extreme values derived from 5-daily gravity field simulations to those from monthly fields, and (ii) show how the weaker spatial

filtering required for MAGIC has a positive influence on the detectability of extremes.For the analysis two different approaches are exploited: One method focuses solely on the stochastic characteristics of the time series in terms of extreme value theory, evaluating the magnitude-frequency relationship of large TWS values by calculating expected return levels of wet extremes. The other approach builds on the fact that a 50-years simulation time series allows to derive statistically meaningful conclusions from directly comparing reference and simulation output on a time series level. We evaluate the time of occurrence of wet extremes on the basis of classification scores assessing correctly and incorrectly identified extreme events.

DOI: 10.5194/gstm2026-69 · https://doi.org/10.5194/gstm2026-69

GSTM2026-70 B.6 · TOWARDS THE OPERATIONAL USE OF SATELLITE GRAVIMETRY · POSTER

From Development to Systematic Production: SAGSA Generates Space-Based Gravimetry Products (L2–L4) from GRACE and GRACE-FO Data for the Scientific Community

Frédéric Thoumieux1, Michaël Ablain2, Alejandro Blazquez1,3, Claude Boniface1, Alexandre Boughanemi2, Marie Bouih2, Sean Bruinsma1,4, Alexandre Couhert1,4, Gerald Dibarboure1, Joel Dorendeu2, Fabien Gachet1,4, Judith Garnier2, Sebastien Gaugain1, Emeric Lavergne1, Hugo Lecomte3,5, Jean-Michel Lemoine1,4, Benoit Meyssignac1,3, Eric Pellereau2, Felix Perosanz1, Julia Pfeffer2, and Franck Reinquin1,4

CNES, Toulouse, 31400, France · MAGELLIUM, Ramonville Saint-Agne, 31520, France · Université de Toulouse, LEGOS (CNES/CNRS/IRD/UT), Toulouse, France · Université de Toulouse, GET (CNES/CNRS/IRD/UT), Toulouse, France · Finnish Geospatial Research Institute, National Land Survey of Finland

For over twenty years, CNES has provided the scientific community with gravimetric solutions ranging from levels L2 to L4.L2: Spherical harmonic and gridded models derived from the GRACE and GRACE-FO missions.

L3: Combinations of these solutions with products from other centers for hydrology, oceanography, and glaciology.

L4: Integration of space-based gravimetry with other observations (altimetry, etc.) to monitor essential climate variables.

The SAGSA project, launched in 2025, is funded by CNES and implemented by Magellium. It automates the production chain from L2 to L3/L4 products, with monthly or bi-annual updates. Products are automatically distributed to the IR DATATERRA data hubs (FORMATERRE, THEIA, ODATIS), leveraging, among others tools, the CNES GEODES portal. By automating these levels, SAGSA establishes an operational framework that will support the future ESA/NASA MAGIC-NGGM mission.

DOI: 10.5194/gstm2026-70 · https://doi.org/10.5194/gstm2026-70

GSTM2026-71 B.4 · HYDROLOGY · ORAL

GRACE/-FO Data Assimilation for Hydrological Applications: From Model-Specific Solutions Towards Broader Use

Anne Springer, Ewerdwalbesloh Yorck, Nitschke Annika, Gerdener Helena, Mielke Christian, Fupeng Li, and Jürgen Kusche

Bonn University, Institute of Geodesy and Geoinformation, Bonn, Germany Bonn University, Institute of Geodesy and Geoinformation, Bonn, Germany

For 20 years now, GRACE and GRACE-FO data have been integrated into hydrological and land-surface models using data assimilation (DA) approaches. Numerous studies have demonstrated the value of GRACE/-FO DA systems for improving estimates of groundwater evolution, enhancing drought monitoring, and providing more accurate flood potential estimates. However, unlike other observation types such as remotely sensed soil moisture, GRACE/-FO observations are rarely assimilated into hydrological and land-surface models on an operational basis. With the prospect of near-real-time observations from upcoming satellite gravimetry missions, the question of how to make GRACE/-FO DA more broadly applicable is becoming increasingly relevant.Currently, there is no single established

method for assimilating GRACE/-FO data into hydrological and land-surface models. While some settings have already reached a certain degree of consensus, for example suitable ensemble-based DA methods and the treatment of GRACE/-FO error covariance matrices, other aspects remain less well established. A key challenge is that GRACE/-FO observes integrated terrestrial water storage, whereas hydrological and land-surface models represent this storage through different components, such as soil moisture, snow, groundwater, and surface water. Assimilating GRACE/-FO data therefore requires determining how the observed TWS information is distributed among the model states. The formulation of the state vector and observation operator plays an important role in this process: the state vector determines which model variables are included in the assimilation, while the observation operator determines how these states are combined to represent the integrated TWS observed by GRACE/-FO. Different formulations of the state vector and observation operator can therefore change how information from the observations is distributed among storage components. Additionally, the representation of model uncertainty can strongly influence the assimilation results. We illustrate these challenges using two different DA systems: one based on the global model WaterGap Global Hydrological Model (WGHM) and one based on the regional high-resolution European land-surface model enCore Community Land Model (eCLM).Our results from eCLM-DA show that the choice of the state vector and the representation of model uncertainty strongly influence the assimilation results. Experiences with WGHM further indicate that some of these challenges are common across modelling systems, while others depend on model structure and resolution. We therefore distinguish between general methodological challenges, model-specific requirements, and computational and operational constraints, and identify areas where further methodological development is needed, including for coupled and multivariate DA systems. These considerations highlight challenges that must be overcome for the broader application of GRACE/-FO DA across modelling systems and regions, including its integration into regional monitoring and early warning systems.

DOI: 10.5194/gstm2026-71 · https://doi.org/10.5194/gstm2026-71

GSTM2026-72 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

Evaluation of the OMP 2025 AOD1B dealiasing product based on the TUGO ocean dynamics numerical model

Fabien Gachet1,2, Alexandre Couhert1,2, Jean-Michel Lemoine1,2, Pascal Gegout1, Florent Lyard3, and Sean Bruinsma1,2

Université de Toulouse, GET (CNES/CNRS/IRD/UT), Toulouse, France · CNES, Toulouse, France · Université de Toulouse, LEGOS (CNES/CNRS/IRD/UT), Toulouse, France

Temporal aliasing of sub-monthly gravimetric signals originating from the Earth's fluid envelope is a major source of error in time-variable gravity field models derived from the GRACE and GRACE Follow-On missions. Minimizing these errors to detect gravimetric signals providing information on solid Earth dynamics and the climate system remains a major challenge.

The GET and LEGOS laboratories in Toulouse have calculated the atmospheric and oceanic contributions to the time-variable gravity field, compiled into the OMP 2025 AOD1B dealiasing products, for the future processing of GRACE and GRACE-FO data.These products rely on an updated and improved version of the TUGO ocean dynamics numerical model and the ECMWF ERA5 3D atmosphere model. These atmospheric and oceanic dealiasing products (OMP 2025) include: (1) time-variable atmospheric gravitational potential anomalies generated from ECMWF ERA5 model levels (including hydrometeors, i.e., specific water content of rain, snow, cloud liquid, and ice), realistically distributed according to altitude above the geoid; and (2) time-variable oceanic gravitational potential anomalies calculated from ocean tides forced by ERA5 ocean surface fields (pressure and wind), with these water height anomalies realistically placed on the geoid. The oceanic response to atmospheric forcing is derived from the TUGO hydrodynamic model without assimilation and the FES2022 ocean tide model (validated by altimetry tides).These new dealiasing products are compared, after GRACE/GRACE-FO data reduction, meaning at the gravity field solution level, with the GFZ AOD1B RL06 and RL07 products officially used in GRACE/GRACE-FO processing. To this end, monthly GRACE gravity field solutions were calculated for 2006 and 2020 using these different dealiasing products, and the resulting monthly solutions are compared to analyze their differences. The analysis of post-fit residual

differences between different dealiasing gravitational potential models, as well as the RMS of quiet zones and differences in monthly GRACE/-FO variable field solutions, will be presented.

Additionally, the new GFZ MIXED2025 ocean tide model is compared to the LEGOS FES2022 model at the gravity field solution level.

DOI: 10.5194/gstm2026-72 · https://doi.org/10.5194/gstm2026-72

GSTM2026-73 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

A detailed description of CNES monthly and 10-day time-variable gravity solutions and a summary of the improvements to be applied to the next reprocessing

Jean-Michel Lemoine1,2, Alexandre Couhert1,2, Fabien Gachet1,2, Sean Bruinsma1, Pascal Gégout2, Franck Reinquin1, Frédéric Thoumieux1, Alexandre Boughanemi3, Eric Pellereau3, and Julia Pfeffer3

CNRS, CNES DTN/CD/GS, Toulouse Cedex 9, France · Université de Toulouse, GET (CNES/CNRS/IRD/UT), Toulouse, France · Magellium, 1 Rue Ariane, 31520 Ramonville-Saint-Agne, France

CNES has been producing solutions of the time-variable gravity field from GRACE and GRACE-FO data for many years. In addition to the unconstrained solutions — which along with many others are submitted to the COST-G international service to derive a combined monthly solution — CNES produces original stabilized solutions at monthly and 10-day time resolutions. These solutions thus differ from both unstabilized spherical harmonic solutions and mascon solutions.This presentation aims to detail the processes used to derive our solutions, specifically the singular value decomposition technique stabilized by a Tikhonov constraint, and the incorporation at the normal equations level of SLR data necessary for the accurate determination of low-degree terms. A comparison of the resulting solutions with spherical harmonic and mascon solutions will be presented. We will also outline efforts underway for a complete reprocessing of GRACE and GRACE-FO data, including the testing of new de-aliasing and ocean tide models, the fixing of GPS integer ambiguities, and the incorporation of instrument and geophysical model variance-covariance matrices required for data processing.

DOI: 10.5194/gstm2026-73 · https://doi.org/10.5194/gstm2026-73

GSTM2026-74 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

COM-TG: A combined mascon product from multiple analyses of GRACE and GRACE Follow-On space gravity data

Rebecca McGirr and Paul Tregoning

The Australian National University, College of Science, Research School of Earth Sciences, Canberra, Australia The Australian National University, College of Science, Research School of Earth Sciences, Canberra, Australia

Currently available GRACE and GRACE Follow-On (GRACE(-FO)) mascon solutions differ due to variations in parameterization, regularization and land-ocean classification, leading to inconsistencies in regional mass change estimates. To reduce dependence on any single processing strategy, we present the first fused mascon solution, COM-TG, derived from four independent GRACE(-FO) products generated by separate analysis centers and combined within a consistent inversion framework. Prior to fusion, inconsistencies in land-ocean signal attribution were addressed using a coastline resolution improvement (CRI) approach. The fused solution is solved on $\sim0~km mascons delimited by coastlines and grounding lines, and is shown to be relatively insensitive to the choice of observation weighting used. It mitigates center-specific regularization assumptions, suppresses noise relative to individual input products and preserves meaningful spatial variability. The resulting product provides a unified representation of GRACE(-FO) mass changes for hydrological, oceanographic and cryospheric applications.

DOI: 10.5194/gstm2026-74 · https://doi.org/10.5194/gstm2026-74

GSTM2026-75 B.2 · CRYOSPHERE · ORAL

Inter-comparison of Greenland-wide annual mass trends estimated with the input-output method and extracted from GRACE level-2 data

Pavel Ditmar1, Weiran Li1, Bert Wouters1, Roland Klees1, Michiel van den Broeke2, and Brice Noël3

Delft University of Technology, Faculty of Civil Engineering and Geosciences, Department of Geoscience and Remote Sensing, Delft, Netherlands · Institute for Marine and Atmospheric Research, Utrecht University, Utrecht, The Netherlands · Department of Geography, University of Liège, Liège, Belgium

Ice mass loss in Greenland attracts considerable public attention because the Greenland Ice Sheet (GrIS) is the largest single contributor to the mass component of global sea-level rise. To project the future evolution of the GrIS over the coming decades, accurate geophysical models are required. In particular, regional climate models, such as the Regional Atmospheric Climate Model RACMO2.3/RACMO2.3p2, are used to estimate the Surface Mass Balance (SMB) of the GrIS and surrounding areas. Data from the GRACE and GRACE Follow-On (GFO) satellite gravity missions provide a primary source of independent information for evaluating modelled mass anomalies integrated over individual GrIS drainage systems and Greenland as a whole.In this study, we compare Greenland-wide estimates of annual mass trends derived from RACMO2.3/2.3p2 with those obtained from GRACE/GFO level-2 data for the period 2003–2023. As input, we use monthly sets of GRACE/GFO-based spherical harmonic coefficients produces at four institutions: the Center of Space Research (CSR) at the University of Texas at Austin; the German Research Centre for Geosciences (GFZ) in Potsdam; the Jet Propulsion Laboratory (JPL) of the California Institute of Technology; and the Institute of Geodesy at Graz University of Technology (ITSG), Austria. The SMB-based estimates from RACMO2.3/2.3p2, combined with information on ice discharge (Mankoff et al, 2020), provide estimates of the total mass trends using the input-output method. We aim to ensure full consistency between the datasets being compared and their processing. Since the spatial resolution of RACMO model is substantially higher than that of GRACE/GFO data, we transform the RACMO-based mass anomalies into the spherical harmonic domain before further processing. In addition, we use fully consistent sets of months for both datasets to mitigate the potential impact of temporal gaps in GRACE/GFO data.The comparison conducted in this study allows us, first and foremost, to draw conclusions about the performance of RACMO2.3/2.3p2. For example, a recent study by Ran et al (Nature, 2024), based on elastic rock deformations measured by GPS during 2009–2015 showed that RACMO2.3p2 may underestimate meltwater runoff in “hot” years and overestimate it in “cold” years. By considering a substantially longer time interval, our study provides an independent evaluation of meltwater runoff modelled by RACMO2.3/2.3p2. At the same time, comparing GRACE/GFO-based estimates from different data processing centres with RACMO-based estimates enables us to assess the accuracy of the considered GRACE/GFO data products in the context of Greenland.

DOI: 10.5194/gstm2026-75 · https://doi.org/10.5194/gstm2026-75

GSTM2026-77 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · POSTER

Preliminary results of the GFZ daily GRACE/GRACE-FO Kalman filter solution

Natalia Panafidina1, Michael Murböck2, Christoph Dahle1, Markus Hauk1, Josefine Wilms1, Karl-Hans Neumayer1, and Frank Flechtner1,2

GFZ Helmholtz Centre for Geosciences, Global Geomonitoring and Gravity Field, Potsdam, Germany · Technical University Berlin, Physical Geodesy

Within the framework of the Research Unit NEROGRAV (New Refined Observations of Climate Change from Spaceborne Gravity Missions), we focus on the computation of daily gravity field solutions using a Kalman filter approach. We use the RL07p GRACE/GRACE-FO daily normal equations, which incorporate refined stochastic modeling of the observations and background models. The stochastic information on the geophysical processes governing submonthly gravity field variations is derived from HISerr time series.

In this study, we present GFZ daily gravity field solutions for the five-year period from 2004 to 2008. To investigate the impact of different processing strategies, several Kalman filter configurations are tested. These include the use of different a priori gravity field models, different approaches to constructing the required hydrological variance-covariance matrix, and the application of a Kalman smoother in comparison with the corresponding raw Kalman filter solutions. The resulting solutions are assessed in terms of their consistency and sensitivity to the different processing choices.

Validation is performed in two ways. First, the GFZ daily solutions are compared with independent daily Kalman filter solutions computed at TU Graz. Second, the daily gravity field solutions are evaluated by comparison with the vertical components of GNSS stations from the global IGS network. This provides an external assessment of the temporal variations captured by the daily gravity field solutions.

DOI: 10.5194/gstm2026-77 · https://doi.org/10.5194/gstm2026-77

GSTM2026-78 B.4 · HYDROLOGY · ORAL

A Physics-Informed Dual-Stream Architecture for Spatiotemporal Downscaling of GRACE/GRACE-FO to Daily Resolution

Metehan Uz1, Kazım Gökhan Atman2, Orhan Akyılmaz3, and C k Shum4

GFZ Helmholtz Centre for Geosciences (Potsdam, Germany), Geodesy, Potsdam, Germany · School of Mathematical Sciences, Queen Mary University of London, London, UK · Dept. of Geomatics Eng., Istanbul Technical University, 34469, Istanbul, Turkey · Space-Analytics.org, USA

We introduce a daily and 25-km spatiotemporally downscaled GRACE/-FO Terrestrial Water Storage Anomaly (TWSA) dataset for the Contiguous United States (CONUS) spanning 2002–2024. To address the resolution gap between coarse monthly satellite observations and daily hydrometeorological dynamics, we developed a physics-informed Dual-Stream Deep Learning (DL) architecture. The framework synchronously trains a monthly physics-constrained estimator by enforcing strict mass conservation via a global mass budget projection and a daily probabilistic estimator that models conditional predictive distributions using a Student-t likelihood to accommodate heavy-tailed GRACE/-FO TWSA residuals. Internal validation demonstrates that the downscaled product preserves the long-term trends and spatial variance of the native GRACE/-FO observations while outperforming standard land surface models that are used as input of DL framework. Crucially, the daily temporal resolution enables the quantification of high-frequency hydrologic extremes that are typically obscured in monthly gravity fields. We extensively validated the daily TWSA dataset against a 22-year catalog of Atmospheric River (AR) events in California and 104 different Tropical Cyclone (TC) events across the eastern CONUS. The downscaled TWSA successfully resolves rapid mass loading with a near-zero phase lag relative to independent in-situ streamflow and precipitation networks. The resulting simulation yields a mass-conserving, physically interpretable daily TWSA product suitable for high-frequency hydrological and climate applications.

DOI: 10.5194/gstm2026-78 · https://doi.org/10.5194/gstm2026-78

GSTM2026-79 B.4 · HYDROLOGY · ORAL

A GRACE-based framework for estimating specific yield at regional scale across India

Maya Raghunath Suryawanshi1, Kuruva Satish Kumar1, Balaram Shaw1, Chethan Varadaganahalli Anandagowda1, Vandana Sukumaran1, Retinder Kour1, Aayushi Kochar1, Shard Chander2, Bhaskar Nikam3, Nagesh Kumar Dasika4, and Bramha Dutt Vishwakarma1,5

Indian Institute of Science, Bengaluru, Interdisciplinary Centre for water Research (ICWaR), Bengaluru, India · Land Hydrology Division, Space Applications Centre, Indian Space Research Organisation; Ahmedabad, India · Indian Institute of Remote Sensing, Indian Space Research

Organisation; Dehradun, India · Department of Civil Engineering, Indian Institute of Science; Bengaluru, India · Centre for Earth Sciences, Indian Institute of Science; Bengaluru, India

Specific yield (Sy) is a fundamental hydrogeological parameter governing groundwater storage and availability in unconfined aquifers. However, its estimation at regional to continental scales remains challenging because of the limited availability of reliable in-situ observations. Existing estimates commonly rely on hydrogeological maps or model-calibrated values, which may involve substantial uncertainties and often do not adequately represent spatial variability. Here, we develop an observation-based framework to estimate spatially varying Sy across India by integrating seasonal in-situ groundwater-level observations with groundwater storage changes derived from GRACE/GRACE-FO observations using the water table fluctuation method. Sy was estimated at the GRACE mascon scale (~3°) for 2003–2022, providing a consistent assessment of regional variability across India. The estimated Sy exhibits distinct spatial patterns, with relatively higher values across north-eastern and parts of peninsular India and lower values across western India. Comparison with reference Sy derived from hydrogeological information shows generally good agreement across several mascons, while differences in selected regions highlight spatial heterogeneity that may not be adequately represented by map-based estimates. Uncertainty analysis based on error propagation indicates that the estimated Sy values largely fall within the uncertainty ranges of the reference estimates. To assess the influence of spatial resolution, Sy was additionally estimated using a downscaled GRACE/GRACE-FO groundwater storage product at 0.5° resolution. The downscaled product provides negligible improvement over the mascon-scale estimates, demonstrating that the GRACE mascon framework is adequate for regional to continental scale Sy estimation. The proposed framework provides an observation constrained approach for characterizing spatial variability in Sy and can support large-scale assessments of groundwater storage and availability. Nevertheless, the estimates remain constrained by the spatial resolution of GRACE observations and the temporal and spatial coverage of groundwater-level measurements, limiting their applicability for local-scale aquifer characterization.

DOI: 10.5194/gstm2026-79 · https://doi.org/10.5194/gstm2026-79

GSTM2026-80 B.4 · HYDROLOGY · POSTER

Assessment of continental water mass balance components 2003--2022 over EURO-CORDEX

Benjamin D. Gutknecht and Jürgen Kusche

Universität Bonn, SFB1502 DETECT, Institut für Geodäsie und Geoinformation, Astronomische, Physikalische und Mathematische Geodäsie (APMG), Bonn, Germany Universität Bonn, SFB1502 DETECT, Institut für Geodäsie und Geoinformation, Astronomische, Physikalische und Mathematische Geodäsie (APMG), Bonn, Germany

Observational and modelled components of the continental water cycle still show considerable variation between available data products. At the same time, however, the research community is trying to identify processes and changes across several domains of the hydrosphere in increasingly more detail.Within the Collaborative Research Centre (CRC) 1502 DETECT, we have developed an analysisframework in order to assess the mass budget consistency of observational and modelled water flux products. We evaluate essential climate variables such as precipitation (P), evaporation from land (E), terrestrial water storage (TWS) change (dTWS/dt), and river discharge (Q) by means of kernel-integrated monthly water mass fluxes in the GRACE/-FO era. Specifically, we focus on the terrestrial water balance equation as in dTWS/dt+Q=P−E, which states that net flux P−E between atmosphere and surface must be balanced by the combination of dTWS/dt and Q.In our major assessment 2003-01/2022-12 over the EURO-CORDEX region, we assessed budget flux components for 30+ major river catchments, including a wide selection of observational and reanalysis products, such as e.g. GPCC, MSWEP, ERA5, GLEAM, GRDC and COST-G (and other GRACE-related products). While individual data products can vary considerably on a per-month basis and in their relative offsets, we find that clear majority (~74%) of catchments show drying P−E behaviour, while the balance-term dTWS/dt+Q shows mostly consistent behaviour (~67%). Notably, out of the basins with a maximum P−E net flux of +25 mm/month, >90% show drying trends in P−E, and >85% show negative trends in the combination of TWS change and river discharge.While dP/dt, across Europe, is heterogeneously distributed spatially and across data

products, E more consistently increased by 1 mm/month per decade. On average, TWS losses increased and discharge declined. In contrast to absolute fluxes, dTWS/dt+Q change appears, on average, ~equally caused by a decline in discharge and TWS; the trends, however, exhibit large uncertainties. In total, a flux budget misclosure remains below 2 mm/month.We again point out that researchers must be aware of the notable variability between available data products and avoid 'cherry-picking' of analysis products.

DOI: 10.5194/gstm2026-80 · https://doi.org/10.5194/gstm2026-80

GSTM2026-81 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Towards a SmallSat-compatible Laser Ranging Instrument for Future Multi-Pair Constellations

Malte Greuner, Joshua Reeder, Max Rohr, and Koch Alexander

DLR Institute for Satellite Geodesy and Inertial Sensing, Hannover, Germany DLR Institute for Satellite Geodesy and Inertial Sensing, Hannover, Germany

Today’s best laser-ranging instruments, such as those on GRACE-FO, GRACE-C and NGGM, reach nanometer-level precision, or have their sensing noise requirement on that level. However, they are large, expensive, and designed for big satellites. The costs for such GRACE-like missions exceed M€ 500, and their gravity fields are usually provided on a monthly level, which is too slow for many hydrologic processes. The MAGIC constellation attempts to increase the temporal resolution, but is limited to ~weekly products. To observe rapid water changes and aid, e.g., in flood-prediction, increased ground-track coverage is required. A cost-effective way to achieve this is with pairs of SmallSats in different orbital planes, carrying compact and comparably cheap payloads.This is the motivation for the miniaturisation of a laser-interferometric ranging payload for SmallSat constellations, with a target size below 8 U, weight below 6 kg and power below 20 W. It seeks ranging precision better than 1 µm/√Hz × NSF(f) between 10^{-3} and 10^{-1} Hz.Instead of using a large optical reference cavity, our miniature LRI will use modulation transfer spectroscopy with a rubidium molecular reference and a 780 nm external-cavity diode laser. The same laser system could be used in a future implementation for a Rb-based accelerometer. The instrument also uses an on-axis optical bench layout to avoid the triple-mirror assembly, and adds satellite communication over the same laser link, enabling data transfer between the two spacecraft, as well as absolute ranging at the centimeter-level.In this presentation, we will report on the design, currently on-going laboratory tests of key enablers for such an instrument as well as the next steps towards the implementation of an Engineering Model. In short, a SmallSat inter-satellite ranging payload turns proven gravity-measurement ideas into a SmallSat-compatible, quantum-aligned payload that could make frequent, low-cost Earth Observation possible - and may later support planetary missions such as the MaQuIs mission around Mars.

DOI: 10.5194/gstm2026-81 · https://doi.org/10.5194/gstm2026-81

GSTM2026-82 B.2 · CRYOSPHERE · ORAL

Precision of Antarctic mass balance estimates from time-variable gravity limited by atmospheric pressure uncertainties

Tyler Sutterley1 and Isabella Velicogna2,3

University of Washington, Applied Physics Laboratory, Polar Science Center, Seattle, United States of America · University of California Irvine, Earth System Science, Irvine, CA, United States of America · Jet Propulsion Laboratory, Caltech, Pasadena, CA, United States of America

We look to understand how uncertainties in atmospheric pressure affect the precision of Antarctic ice sheet mass balance estimates derived from time-variable gravity measurements using an ensemble of reanalysis outputs along with in-situ observations of surface pressure. Before the formulation of monthly spherical harmonic gravitational fields from GRACE and GRACE Follow-On, sub-monthly atmospheric variations are removed using the "de-aliasing" products computed from ERA reanalyses and ECMWF operational forecasts and processed by the German Research Centre for Geosciences (GFZ). While this helps remove the majority of the atmospheric signal globally, the quality of

atmospheric forecasts and analyses is typically lower over Antarctica due to the scarcity of available data and the strong topographic contrasts. As uncertainties in the dealiasing fields directly map into errors in the spherical harmonic fields, improving the characterization of atmospheric circulation over the poles will help improve the month-by-month precision of Antarctic mass change estimates. Here, we estimate the impacts of atmospheric pressure uncertainty on ice sheet mass balance at both continental and regional scales using four independently derived reanalyses (ERA5, MERRA-2, JRA-3Q, and NCEP-CFSR). We find that equivalent atmospheric pressure uncertainties of 1 millibar over the entirety of Antarctica will result in a mass uncertainty of over 100 gigatonnes. From our reanalysis ensemble, we find average monthly uncertainties of 0.6–1.1 millibars (71–132 gigatonnes) over the continent, likely limiting our present-day mass change estimates over short timescales. Additionally, we calculate the atmospheric loading effects in Antarctica under different scenarios related to the structure of the Earth's atmosphere and Earth's geometry. We find that for the purposes of deriving atmospheric pressure uncertainties, simple two-dimensional formulations are likely adequate.

DOI: 10.5194/gstm2026-82 · https://doi.org/10.5194/gstm2026-82

GSTM2026-83 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

Status of the GRATTIS Mission

John Conklin1 and the The GRATTIS Team*

University of Florida, Precision Space Systems Laboratory, Mechanical and Aerospace Engineering, United States of America *A full list of authors appears at the end of the abstract · University of Florida, Precision Space Systems Laboratory, Mechanical and Aerospace Engineering, United States of America *A full list of authors appears at the end of the abstracts

The Gravitational Reference Advanced Technology Test In Space (GRATTIS) mission will demonstrate the functionality and sensitivity performance of the Simplified Gravitational Reference Sensor (S-GRS), an ultra-precise inertial sensor for future Earth geodesy missions. These sensors are used to measure or compensate for all non-gravitational accelerations of the host spacecraft so that they can be removed in the data analysis to recover spacecraft motion due to Earth’s gravity field. Low-low satellite-to-satellite tracking missions like GRACE-FO that utilize laser ranging are technologically limited by the acceleration noise performance of their accelerometers, as well as temporal aliasing associated with Earth’s dynamic gravity field.The S-GRS concept is a simplified version of the LISA Pathfinder GRS. It consists of a free-falling cubic test mass (TM) inside an electrode housing that senses the position and orientation of the TM and electrostatically applies forces and torques to keep it centered at a nanometer-level. The applied forces and torques required are also used to determine the non-gravitational forces acting on the host spacecraft, as well as the spacecraft’s angular acceleration. Improvements incorporated into the S-GRS include removing the small grounding wire used in the GRACE accelerometers, which limits performance, and replacing it with a UV LED-based charge management system, increasing the mass of the sensor’s TM, and increasing the gap between the TM and its electrode housing.GRATTIS will fly two identical S-GRS mounted next to one another near the center of mass of a 235 kg ESPA-class commercial microsatellite. The six-axis acceleration measurement capability of the S-GRS allows precision measurement of the spacecraft drag-induced translational acceleration, as well as the residual angular acceleration of the nominally inertially-pointed bus. By combining the outputs of each sensor and with the known relative position of the two TMs, we can recover the acceleration sensitivity (noise floor) of the S-GRS. Our mission goal is to demonstrate acceleration noise performance of ≤10−10 m/s2Hz1/2.The PI and science team is led by the University of Florida (UF) and includes relevant experts from Texas A&M University (TAMU) and CrossTrac Engineering. The S-GRS mechanical sensor heads are provided by BAE Space & Mission Systems, while the S-GRS electronics units are provided by Fibertek, Inc. CrossTrac Engineering provides the S-GRS software and program management. The flight S-GRS payload has been integrated and tested at the University of Florida in August-September 2026. The payload was then delivered to the spacecraft provider (Apex Space) and is in the final stages of integration and test with the spacecraft. The integrated spacecraft will be delivered to SpaceX in late 2026 for integration with the launch vehicle. GRATTIS is manifested on SpaceX Transporter-19 launching from Vandenburg Space Force Base into a 561 km sun-synchronous orbit no earlier than February 1, 2027. This presentation will describe the GRATIS technology demonstration mission, current status, and expected results.

DOI: 10.5194/gstm2026-83 · https://doi.org/10.5194/gstm2026-83

GSTM2026-84 A.1 · GRACE-FO PROJECT STATUS · ORAL

GFZ SDS Level-2/-3 Overview

Christoph Dahle1, Markus Hauk1, Michael Murböck1,2, Natalia Panafidina1, Josefine Wilms1, and Annette Eicker1

GFZ Helmholtz Centre for Geosciences, Geodesy, Potsdam, Germany · Technische Universität Berlin, Institute of Geodesy, Berlin, Germany

This presentation will give an overview of GFZ's activities within the GRACE-FO Science Data System (SDS) including the current status of Level-2 and Level-3 products. It will also provide a high-level summary of the ongoing RL07 reprocessing campaign. First GFZ RL07 Level-2 products are expected to be published shortly after the GSTM2026; compared to the currently operational GFZ RL06 time series, they show a significant reduction of noise (> 30 %) and more realistic formal error estimates.

DOI: 10.5194/gstm2026-84 · https://doi.org/10.5194/gstm2026-84

GSTM2026-85 B.1 · SOLID EARTH SCIENCES · ORAL

Mantle Rheology Across Tidal to Decadal Timescales

Lambert Caron, Erik Ivins, Surendra Adhikari, and Erin Hightower

Jet Propulsion Laboratory, California Institute of Technology Jet Propulsion Laboratory, California Institute of Technology

Long-period deformation of the solid Earth provides a unique window into mantle rheology between seismic and glacial-isostatic-adjustment (GIA) timescales, where constitutive behavior remains poorly constrained. Laboratory experiments increasingly support broad transient anelastic relaxation, while geodynamic studies disfavor the extremely low steady-state viscosities that would otherwise be required to explain some long-period observations. The Extended Burgers Material (EBM) offers a linear viscoelastic framework that captures this transient relaxation while asymptotically approaching Maxwell behavior at long timescales.Using newly developed Love-number capabilities in the Ice-sheet and Sea-level System Model (ISSM), we compile and model complex degree-2 tidal Love-number constraints k, h, and l from satellite laser ranging, satellite gravity, Earth-rotation observations, GNSS, and DORIS over periods from 12 hours to 18.6 years. We first test whether the observations can be explained by radial steady-state viscosity structure alone using a large ensemble of flexible layered Maxwell models. These models reproduce much of the gravitational and radial response, but systematically fail to reproduce the real horizontal response, even when very low viscosities are permitted in the lowermost mantle or introduced in the transition zone and asthenosphere.We then explore EBM rheologies with distinct upper- and lower-mantle properties. These models reproduce the joint frequency dependence of Love numbers significantly better while retaining steady-state viscosities compatible with GIA constraints. The results point to an important role for transient lower-mantle relaxation, although correlations between relaxation amplitude and timescale, and the long-period cutoff of the transient response, remain incompletely resolved. Better constraints on this behavior are directly relevant to GRACE and GRACE-FO, as well as GNSS, SLR, DORIS, altimetry, and other remote-sensing observations that depend on accurate modeling of solid-Earth deformation across overlapping timescales.© 2026 California Institute of Technology. Government Sponsorship Acknowledged.

DOI: 10.5194/gstm2026-85 · https://doi.org/10.5194/gstm2026-85

GSTM2026-86 A.2 · ANALYSIS TECHNIQUES & INTER-COMPARISONS · ORAL

GRACE Follow-On LRI measurements of linear accelerations produced by

attitude control thrust firings

Shin-Chan Han1 and George Pescod2,1

Geodetic Science, The Ohio State University, Columbus, USA · Physics, California State University, Fullerton, USA

The attitude of the GRACE Follow-On satellites is controlled by six pairs of attitude control thrusters. An ideal thruster system would produce a change only in angular momentum. The imperfect system, however, also produces undesired linear accelerations that must be removed for orbit analysis and gravity recovery. On-orbit calibration is performed to determine the linear acceleration response to each of the six thrust events for both satellites. These thrust calibration models are applied to the ACC1B data product and its variants. As is standard for L1B data products, a CRN low-pass filter is applied with a cutoff frequency of 35 mHz for the case of the ACC1B datasets. This cutoff frequency was chosen for orbit and gravity analysis using KBR1B from the beginning of the GRACE mission, considering the balance between dampening gravitational signals and increasing KBR thermal noises with frequency. We found spurious acceleration signals, most clearly visible in the form of range acceleration computed from LRI1B data and a prefit dynamic orbit. These spurious accelerations were in fact LRI measurements of thrust events — all six cases — appearing in a frequency band well above 35 mHz. We discuss how the LRI measures thruster firings, how we model and remove them, and why this modeling matters for global gravity recovery.

DOI: 10.5194/gstm2026-86 · https://doi.org/10.5194/gstm2026-86

GSTM2026-87 B.4 · HYDROLOGY · ORAL

Attribution of spatially refined terrestrial water storage trends to human activities

Mary Michael ONeill1,2, Matt Rodell3, and Bryant Loomis4

Hydrological Sciences Laboratory, NASA Goddard Space Flight Center, United States of America · Earth System Science Interdisciplinary Center, University of Maryland, United States of America · Earth Sciences Division, NASA Goddard Space Flight Center, United States of America · Geodesy and Geophysics Laboratory, NASA Goddard Space Flight Center, United States of America

Human activities are reshaping Earth's freshwater stores, but the spatial resolution of satellite gravimetry has limited its utility in resolving these changes at scales relevant to water management. Here we use a "stacked" regression of 23 years of GRACE and GRACE Follow-On Level-1B intersatellite range observations to map global terrestrial water storage (TWS) trends at finer spatial resolution than conventional monthly mascon products. The resulting trend map resolves features at approximately 12,500 km² and retains substantially more of the underlying mass signal. We combine this spatially refined solution with uncertainty analysis, spatial statistical methods, and independent datasets describing land and water use, climate variability, and glacier mass loss, to investigate the drivers of observed TWS trends. We identify 40 regions where TWS trends are significantly associated with anthropogenic activities, including groundwater and surface-water irrigation, rainfed agriculture, deforestation, and reservoir impoundment. Regional TWS trend magnitudes are, on average, larger than those derived from conventional monthly mascon products and those reported in previous studies, particularly in over-exploited aquifers. The higher-resolution solution also reveals that localized freshwater gains associated with rainfed agriculture, surface-water irrigation, and reservoir filling, and losses from groundwater pumping and agriculture, are globally pervasive, with anthropogenic independent variables explaining a significant portion of TWS trend spatial variability in every non-polar continent except Australia. These results provide a detailed picture of global mass change and demonstrate the potential of improved satellite gravimetry processing to reveal human influences on freshwater storage.

DOI: 10.5194/gstm2026-87 · https://doi.org/10.5194/gstm2026-87

GSTM2026-88 A.3 · GRACE-C, NGGM, AND FUTURE MISSIONS · ORAL

GRACE-C Implementation Status - Update

Felix Landerer1, David Wiese1, William Klipstein1, Sebastian Fischer2, Himanshu Save3, and Chris McCullough1

Jet Propulsion Laboratory / California Institute of Technology, Pasadena, USA · Germany Space Agency at DLR, Department Earth Observation,

Germany · University of Texas at Austin, Center for Space Research, United States

GRACE-Continuity (GRACE-C) is a joint NASA (US) & DLR (Germany) mission, scheduled for launch in Dec-2028. It will continue the essential data record of Earth system gravity and mass change initiated by the original GRACE mission (2002-2017) and currently enabled by the GRACE-FO mission (2018 – present). The GRACE-series provides foundational observations of monthly to decadal global mass changes and transports in the Earth system derived from temporal variations in the Earth's gravity field. These data are indispensable for accurately measuring the evolving global water cycle, including ocean dynamics, polar ice mass changes, and near-surface and global ground water changes. Mass Change data enable the monitoring of flood potentials as well as droughts by tracking freshwater availability, groundwater and aquifer volume changes. These observations and integrated tools built around them have become indispensable for 'water intelligence' to inform data-driven water-management.We will present an update on the implementation status of GRACE-C: the mission is well into its assembly, integration and testing phase. Key science instruments like the laser ranging interferometers, accelerometers, GNSS-receivers and star cameras are undergoing testing, and are being integrated on the two GRACE-C satellites. The GRACE-C SDS team is continuously assessing and verifying the measurement system performance throughout the flight system integration phase. The mission is on schedule to launch in late 2028. We will provide an assessment of the expected science performance, as well as plans for low- and high-level data products and algorithms to support optimal science and application mission return.

DOI: 10.5194/gstm2026-88 · https://doi.org/10.5194/gstm2026-88

GSTM2026-90 A.1 · GRACE-FO PROJECT STATUS · ORAL

GRACE and GRACE-FO Level-1 Data Processing Status

Christopher McCullough, Willy Bertiger, Sung Byun, Eugene Fahnestock, Christopher Finch, Felix Landerer, Mark Miller, Meegyeong Paik, Athina Peidou, and Christopher Volk

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, USA Jet Propulsion Laboratory, California Institute of Technology, Pasadena, USA

The Gravity Recovery and Climate Experiment Follow-On (GRACE-FO), launched in May 2018, provides invaluable information about mass change in the Earth system, continuing the legacy of GRACE. Fundamental requirements for successful mass change recovery are precise orbit determination and inter-satellite ranging, determination of the relative clock alignment of the ultra-stable oscillators (USOs), precise attitude determination, and accelerometry. NASA/Caltech Jet Propulsion Laboratory is the official Level-1 data processing and analysis center and provides weekly Level-1A/B data updates. Here we present analysis of the aforementioned GRACE-FO sensor data, status of the accelerometer transplant calibrations, as the basis for Level-2 release RL06.3, and a discussion of the overall GRACE-FO measurement system performance through Solar cycle 25. We also provide an update on current GRACE/GRACE-FO reprocessing efforts toward a future unified Release-07 data record.

The research presented in this abstract has been carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.

©2026 California Institute of Technology. Government sponsorship acknowledged.

DOI: 10.5194/gstm2026-90 · https://doi.org/10.5194/gstm2026-90

GSTM2026-91 A.1 · GRACE-FO PROJECT STATUS · ORAL

GRACE-FO Mission Operations Report

Krzysztof Snopek

GFZ Helmholtz Centre for Geosciences, Germany GFZ Helmholtz Centre for Geosciences, Germany

In this presentation the status of the GRACE-FO mission and the flight segment will be given.

DOI: 10.5194/gstm2026-91 · https://doi.org/10.5194/gstm2026-91

GSTM2026-92 B.4 · HYDROLOGY · ORAL

GRACE/GRACE-FO Observations Reveal Long-Term Groundwater Depletion and Freshwater Storage Losses in the Colorado River and Rio Grande-Bravo Basins

karem Abdelmohsen1, James S. Famiglietti1, and Brian D. Richter2

Arizona State University, School of Sustainability , Tempe, United States of America · Sustainable Waters, Abiquiu, USA

GRACE and GRACE Follow-On (GRACE-FO) satellite observations provide a unique capability for quantifying long-term changes in terrestrial water storage and assessing groundwater depletion across large river basins where in situ observations are spatially limited or inconsistent. Here, we use GRACE/GRACE-FO observations together with land surface models, reservoir storage records, and water-use information to evaluate freshwater storage changes and their partitioning in two major transboundary river systems of the southwestern United States and northern Mexico: the Colorado River Basin and the Rio Grande-Bravo Basin.GRACE/GRACE-FO observations reveal persistent long-term water storage declines across both basins, with groundwater depletion representing a major component of the observed losses. In the Colorado River Basin, total water storage declined by 52.2 km^3, with groundwater accounting for approximately 65% of the total loss, indicating that groundwater depletion has exceeded surface reservoir declines as the dominant source of freshwater loss. In the Rio Grande-Bravo Basin, groundwater depletion averages 2.7 km^3 yr^{-1}, while 52% of direct anthropogenic water consumption is unsustainable and irrigated agriculture accounts for 87% of direct consumptive use. These results demonstrate the increasing reliance on groundwater to compensate for declining renewable surface-water supplies.The analysis highlights the value of GRACE/GRACE-FO for interpreting long-term hydrological trends and separating groundwater storage changes from other terrestrial water storage components. Combining satellite gravimetry with complementary hydrological datasets provides a consistent framework for evaluating basin-scale freshwater storage variations, identifying regions of persistent groundwater depletion, and improving understanding of the processes driving water storage decline in heavily managed and water-stressed river basins.

DOI: 10.5194/gstm2026-92 · https://doi.org/10.5194/gstm2026-92

GSTM2026-95 A.1 · GRACE-FO PROJECT STATUS · ORAL

TN-14: Validation efforts and impacts of AOD1B RL07

Bryant Loomis, Kenneth Rachlin, and Ben Griffin

NASA GSFC, Greenbelt, United States of America NASA GSFC, Greenbelt, United States of America

We present the status of the low degree gravity coefficients C20 and C30 derived from satellite laser ranging and distributed as Technical Note 14. This includes comparisons to solutions from other centers, our group's new long-term C20 solution that is independent of the background gravity model, and new length-of-day analysis. We also present and discuss the impacts of updating the AOD1B model from RL06 to RL07 on the C20 and C30 estimates.

DOI: 10.5194/gstm2026-95 · https://doi.org/10.5194/gstm2026-95

GSTM2026-96 A.1 · GRACE-FO PROJECT STATUS · ORAL

Status of AOD1B

Linus Shihora, Laura Jensen, Robert Dill, and Henryk Dobslaw

GFZ Helmholtz Centre for Geosciences, 1.3 Earth System Modelling, Potsdam, Germany GFZ Helmholtz Centre for Geosciences, 1.3 Earth System Modelling, Potsdam, Germany

The Atmosphere and Ocean non-tidal De-aliasing Level-1B (AOD1B) product is widely used in satellite gravimetry to correct for transient effects of atmosphere-ocean mass variability that would otherwise alias into monthly-mean global gravity fields. The most recent release is based on the global ERA5 reanalysis and ECMWF operational data

together with simulations from the general ocean circulation model MPIOM consistently forced with fields of the same atmospheric data-set.

In this presentation we give a short overview of the current production status of AOD1B as well as current and future work on hydrological de-aliasing datasets.

DOI: 10.5194/gstm2026-96 · https://doi.org/10.5194/gstm2026-96

GSTM2026-97 B.6 · TOWARDS THE OPERATIONAL USE OF SATELLITE GRAVIMETRY · ORAL

From GRACE Science to Climate Service: Operational TWSA and GWSC Products in the Copernicus Climate Change Service (C3S)

Julian Haas1, Eva Boergens2, Christoph Dahle3, Henryk Dobslaw2, Wouter Dorigo4, Inés Duissaillant7, Annette Eicker3, Frank Flechtner3, Miriam Kosmale5, Johanna Lems4, Kari Luojus5, Wolfgang Preimesberger4, Ehsan Sharifi6, Michael Zemp7, and Andreas Güntner1,8

GFZ Potsdam, Section 4.4 - Hydrology, Potsdam, Germany · GFZ Potsdam, Section 1.3 - Earth System Modelling, Potsdam, Germany · GFZ Potsdam, Section 1.2 - Global Geomonitoring and Gravity Field, Potsdam, Germany · TU Wien, Department of Geodesy and Geoinformation, Vienna, Austria · Finnish Meteorological Insitute, Helsinki, Finland · KIT, Institute of Meteorology and Climate Research, Karlsruhe, Germany · Universität Zürich, Department of Geography, Zurich, Switzerland · University of Potsdam, Institute of Environmental Science and Geography, Potsdam, Germany

Over nearly 25 years, GRACE and GRACE-FO have matured from a novel observing system into an operational-grade data source, with terrestrial water storage anomaly (TWSA) and, derived from this, groundwater storage change (GWSC) products now feeding directly into drought, flood, and water-availability assessments. Despite this maturity, systematic translation of GRACE-based TWSA and GWSC into a sustained, standards-compliant climate service has so far been missing. This gap has limited uptake by climate adaptation and water-management communities which rely on Essential Climate Variable (ECV)-type products with defined uncertainty and provenance.

In this contribution, we present the operationalisation of GRACE-based hydrological science into a climate service through the introduction of a new Essential Climate Variable (ECV) Service, “Terrestrial Water Storage and Groundwater”, within the Copernicus Climate Change Service (C3S). The service delivers Climate Data Records (CDRs) for TWSA and GWSC, built to C3S CDR requirements for homogeneity, traceability, and long-term consistency. The datasets, together with a comprehensive Data Documentation Package following C3S standards, are published on the Climate Data Store (CDS).

A key challenge in this operationalisation is quality characterisation in the near-absence of suitable in-situ reference data at GRACE's native spatial and temporal resolution. We address this with a dedicated quality assessment framework combining internal consistency checks (e.g., mass conservation, basin-scale closure), inter-comparison with global hydrological models and land-surface/reanalysis products, and the explicit documentation of known limitations and fitness-for-purpose. Additionally, TWSA is compared against independent mascon and spherical-harmonic solutions and GWSC against regionalised in-situ measurements from groundwater boreholes.

As the C3S strives towards more accessible information for the public and decision makers, both products contribute to Climate Intelligence efforts through a variety of visualisations across different time scales, which will also be presented here.

DOI: 10.5194/gstm2026-97 · https://doi.org/10.5194/gstm2026-97

GSTM2026-99 A.1 · GRACE-FO PROJECT STATUS · ORAL

CSR GRACE and GRACE-FO Level-2/-3 Overview

Himanshu Save, Chaoyang Zhang, Byron Tapley, Srinivas Bettadpur, Nicholas Childress, Mark Tamisiea, Peter Nagel, Nadege

Pie, Benjamin Krichman, Geethu Jacob, Zhigui Kang, Steven Poole, and John Ries

The University of Texas at Austin, Center for Space Research The University of Texas at Austin, Center for Space Research

We present the GRACE and GRACE-FO Level-2 and Level-3 products generated at CSR, evaluating their quality using standard metrics. We also report on the RL07 reprocessing of GRACE and GRACE-FO data, which incorporates updated Level-1B observations, reference frames, background models, instrument calibrations, and processing strategies. New CSR mascon solutions are being developed within the same RL07 framework. The RL07 solutions reduce the RMS of the noise over the ocean by more than 35% compared to RL06.3, improving solution stability and the effective spatial resolution of mass-change estimates.

DOI: 10.5194/gstm2026-99 · https://doi.org/10.5194/gstm2026-99

GSTM2026-101 A.1 · GRACE-FO PROJECT STATUS · ORAL

GRACE-FO: Science Management Report

Himanshu Save

The University of Texas at Austin, Center for Space Research The University of Texas at Austin, Center for Space Research

The presentation will summarize recent operational activities and future plans aimed at maintaining the high quality of science products.

DOI: 10.5194/gstm2026-101 · https://doi.org/10.5194/gstm2026-101

GSTM2026-103 A.1 · GRACE-FO PROJECT STATUS · ORAL

GRACE-FO: mission status, science performance and outlook

Felix Landerer1, Annette Eicker2, Himanshu Save3, Christopher Mcculloughı, Christoph Dahle2, Mona Witkowski1, Krzysztof Snopek2, John T. Reager1, Athina Peidou1, and Srinivas Bettadpur3

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA · GFZ Helmholtz Centre for Geosciences, Potsdam, Germany · Center for Space Research, University of Texas, Austin, USA

Since 2018, the GRACE Follow-On satellite mission, a partnership between NASA (US) and GFZ (Germany), is continuing the unique essential climate data record of mass change in the Earth system initiated in 2002 by the GRACE mission (2002-2017). The combined GRACE & GRACE-FO data records now span over 22 years and provide foundational observations of monthly to decadal global mass changes and transports in the Earth system derived from temporal variations in the Earth's gravity field. These observations have become indispensable for climate-related studies that enable process understanding of the evolving global water cycle, including ocean dynamics, polar ice mass changes, and near-surface and global ground water changes.

We will present recent GRACE/GRACE-FO science and applications highlights, status of the data processing, calibration approaches, performance of last year's science data, and discuss the GRACE-FO mission operations plan to collect and provide high-quality science data, aiming for overlap with the upcoming NASA/DLR GRACE-Continuity mission, targeted to launch in late 2028.

DOI: 10.5194/gstm2026-103 · https://doi.org/10.5194/gstm2026-103